

#1 · Professional Practice and Legal Communication

What percentage of the total AEMCA exam score does the professional practice and legal communication domain carry?

Answer

It is worth seventeen percent of the total score, making it the largest single domain on the exam.

#2 · Professional Practice and Legal Communication

What does the acronym EHRAB stand for and where does it sit within Ontario's government?

Answer

EHRAB stands for the Emergency Health Regulatory and Accountability Branch, and it sits within the Ontario Ministry of Health.

#3 · Professional Practice and Legal Communication

What authority does EHRAB have over paramedic certification and ambulance services?

Answer

EHRAB sets provincial standards for paramedic certification, ambulance service operation, and equipment, issues the Advanced Emergency Medical Care Assistant certificate, and has authority to investigate complaints, suspend certification, and audit ambulance services for compliance.

#4 · Professional Practice and Legal Communication

What role does a Base Hospital play in Ontario's EMS system?

Answer

A Base Hospital provides medical oversight for paramedics, and its medical director is legally responsible for the medical acts a paramedic performs under delegation.

#5 · Professional Practice and Legal Communication

Why is a patch call to a Base Hospital considered a legal document?

Answer

What is said and what is authorized during a patch call becomes part of the patient record and can be reviewed later, so it carries the same legal weight as other documentation.

#6 · Professional Practice and Legal Communication

Who is responsible for delivering land ambulance service under the Ambulance Act, and what does that responsibility include?

Answer

Upper-tier municipalities, meaning counties and regional municipalities, are responsible for delivering land ambulance service in their geographic area, and they remain accountable for response times and service levels even if they contract the service to another provider.

#7 · Professional Practice and Legal Communication

Describe the three-way relationship that structures Ontario's EMS system.

Answer

EHRAB sets the provincial rules, the Base Hospital owns medical oversight, and the upper-tier municipality owns operational delivery of the ambulance service.

#8 · Professional Practice and Legal Communication

Do professional paramedic associations have regulatory authority over individual paramedics?

Answer

No. Their role is advocacy, professional development, and representing members' interests, not regulation, and association membership should not be confused with certification status.

#9 · Professional Practice and Legal Communication

Why do personal hygiene and a properly worn uniform matter beyond appearance?

Answer

They affect infection control and they affect a patient's trust in the care being provided, so they are functional standards, not just cosmetic details.

#10 · Professional Practice and Legal Communication

What kind of language is expected on scene when speaking with a patient?

Answer

Language should be plain, respectful, and free of jargon the patient cannot understand, and profanity or dismissive language toward a patient, family member, or bystander is a conduct violation regardless of how stressful the call becomes.

#11 · Professional Practice and Legal Communication

A paramedic personally disapproves of a patient's lifestyle. What standard of care must they provide?

Answer

The same standard of care as any other patient, regardless of the paramedic's personal opinion of the patient's lifestyle, appearance, or reason for calling.

#12 · Professional Practice and Legal Communication

Give two examples of inappropriate patient interactions described in the chapter.

Answer

Unnecessary physical contact and comments about a patient's body or choices that are not clinically relevant are both inappropriate, along with any behaviour that could reasonably be seen as flirtatious or intimidating.

#13 · Professional Practice and Legal Communication

Why does the conduct standard exist regarding a paramedic's power over a patient?

Answer

A paramedic is in a position of power over a vulnerable person, and the standard of conduct exists specifically to protect that person from misuse of that power.

#14 · Professional Practice and Legal Communication

What is the purpose of a quality assurance program in EMS?

Answer

Quality assurance programs review call records, patch calls, and patient outcomes to identify where care can improve, and paramedics should treat this review as a tool for improvement rather than a threat.

#15 · Professional Practice and Legal Communication

What does patient advocacy require of a paramedic when a family member's wishes conflict with the patient's own?

Answer

Patient advocacy means speaking up for a patient's needs, including refusing to simply follow a family member's wishes when they conflict with a competent patient's own wishes.

#16 · Professional Practice and Legal Communication

What are community paramedicine programs and why do they exist?

Answer

They extend paramedics into non-emergency roles such as home visits for frequent callers, chronic disease monitoring, and falls prevention, and they exist because a large share of ambulance calls are not true emergencies.

#17 · Professional Practice and Legal Communication

Is continuing professional development a one-time requirement met at graduation?

Answer

No. Continuing professional development and reflective practice are ongoing obligations throughout a paramedic's career, not one-time requirements.

#18 · Professional Practice and Legal Communication

What does reflective practice mean for a paramedic?

Answer

Reflective practice means honestly reviewing one's own calls, including the ones that went poorly, and identifying what would be done differently.

#19 · Professional Practice and Legal Communication

What is evidence-based practice, according to the chapter?

Answer

Evidence-based practice means that clinical decisions and protocols are grounded in research and outcome data, not tradition or personal preference, and paramedics should understand why a directive says what it says.

#20 · Professional Practice and Legal Communication

Name three techniques used to de-escalate a patient in acute psychological crisis.

Answer

A calm tone of voice, non-threatening body positioning, and giving the patient room rather than crowding them are all de-escalation techniques described in the chapter.

#21 · Professional Practice and Legal Communication

Why should a paramedic avoid arguing over the content of a patient's delusion during a crisis call?

Answer

Arguing over the content of a delusion or the accuracy of a complaint increases the chance that a verbal crisis becomes physical, so it should be avoided in favour of active listening.

#22 · Professional Practice and Legal Communication

Why should a paramedic never make a promise they cannot keep to calm a distressed patient?

Answer

A broken promise destroys the patient's trust and can escalate the situation later, so paramedics should avoid promises they are not certain they can fulfill.

#23 · Professional Practice and Legal Communication

What does the abbreviation CIS refer to in EMS, and what does it describe?

Answer

CIS stands for critical incident stress, which describes the psychological impact of exposure to traumatic calls, including pediatric deaths, multiple casualty incidents, and calls involving violence toward the paramedic.

#24 · Professional Practice and Legal Communication

What is CISM and what does it include?

Answer

CISM, Critical Incident Stress Management, is a structured peer support system that includes defusing sessions shortly after a call and more formal debriefing for larger incidents.

#25 · Professional Practice and Legal Communication

How does the role of a peer support worker differ from a mental health professional's?

Answer

Peer support workers are colleagues trained to provide a first line of confidential support, and their role is distinct from a mental health professional's, though referral to professional support is part of a healthy CISM system.

#26 · Professional Practice and Legal Communication

Why is it important for a paramedic to recognize cumulative stress in themselves and colleagues, not just acute reactions to a single bad call?

Answer

Because stress effects can be chronic and build over time, recognizing cumulative, chronic effects rather than only the acute reaction to one traumatic call is part of professional self-care.

#27 · Professional Practice and Legal Communication

What are open-ended questions and reflective listening examples of?

Answer

They are examples of therapeutic communication techniques, along with appropriate silence that gives a patient time to answer.

#28 · Professional Practice and Legal Communication

What is a limitation of relying only on closed questions during a patient history?

Answer

Closed questions gather specific facts quickly, but relying only on them can miss important information a patient would have volunteered given the chance.

#29 · Professional Practice and Legal Communication

What elements should a standard bedside verbal report to a receiving nurse or physician cover?

Answer

It should cover patient identification, chief complaint, relevant history, findings, treatment given, and response to treatment, in an organized structure that does not waste the receiving provider's time.

#30 · Professional Practice and Legal Communication

Why does radio communication follow a disciplined structure in EMS?

Answer

Radio channels are shared and a poor transmission can be misheard with real consequences, so speaking clearly, using plain language instead of unofficial slang, and confirming a message was received and understood are essential disciplines.

#31 · Professional Practice and Legal Communication

In what order should information be given during an organized patch call to a Base Hospital physician?

Answer

The patient's age, chief complaint, relevant history, vital signs, and the specific request or question should be given in that order so the physician can respond efficiently.

#32 · Professional Practice and Legal Communication

Why does a patch call require special attention compared to other communication?

Answer

The Base Hospital physician is making a decision based entirely on what the paramedic tells them, so the information must be organized and accurate.

#33 · Professional Practice and Legal Communication

Name the four ethical principles that guide difficult prehospital decisions.

Answer

Autonomy, beneficence, non-maleficence, and justice are the four ethical principles that guide difficult decisions in the field.

#34 · Professional Practice and Legal Communication

What does the ethical principle of autonomy protect?

Answer

Autonomy respects a capable patient's right to make their own decisions about their own body, including the right to refuse treatment or transport even when a paramedic disagrees with that choice.

#35 · Professional Practice and Legal Communication

What is the difference between beneficence and non-maleficence?

Answer

Beneficence is the obligation to act in the patient's best interest, while non-maleficence is the obligation to avoid causing harm, which sometimes means choosing not to perform an intervention when the risk outweighs the benefit.

#36 · Professional Practice and Legal Communication

What does the ethical principle of justice mean in prehospital care?

Answer

Justice means treating patients fairly and providing the same standard of care regardless of who they are or their ability to pay, and it also shapes decisions like resource allocation during a multiple casualty incident.

#37 · Professional Practice and Legal Communication

A capable patient refuses transport after a seizure. Which two ethical principles are in tension, and what is the correct approach?

Answer

The patient's autonomy conflicts with the paramedic's instinct toward beneficence, and the correct approach respects a capable, informed refusal while documenting that the risks were explained.

#38 · Professional Practice and Legal Communication

What Ontario legislation governs consent to treatment?

Answer

The Health Care Consent Act, 1996, governs consent to treatment in Ontario.

#39 · Professional Practice and Legal Communication

What makes a patient capable under the Health Care Consent Act?

Answer

A capable patient is one who understands the information relevant to a decision and appreciates the reasonably foreseeable consequences of that decision.

#40 · Professional Practice and Legal Communication

Can a capable patient refuse life-saving treatment in Ontario?

Answer

Yes. A capable patient has the right to give or refuse consent to any treatment, including life-saving treatment.

#41 · Professional Practice and Legal Communication

What four elements must informed consent include?

Answer

Informed consent requires the patient understand the nature of the proposed treatment, its expected benefits, its material risks, and the alternatives, including the alternative of no treatment.

#42 · Professional Practice and Legal Communication

When does implied consent apply?

Answer

Implied consent applies when an unconscious or otherwise incapable patient requires emergency treatment and no substitute decision-maker is available, since the law presumes a reasonable person would consent to necessary emergency care.

#43 · Professional Practice and Legal Communication

What is a substitute decision-maker and when does one act?

Answer

A substitute decision-maker acts on behalf of a patient who is not capable of consenting, following a hierarchy set out in the Health Care Consent Act.

#44 · Professional Practice and Legal Communication

List the hierarchy of substitute decision-makers set out in the Health Care Consent Act.

Answer

The hierarchy starts with a court-appointed guardian or a person named in a power of attorney for personal care, then moves through spouse or partner, parent or child, sibling, and other relative, in that defined order.

#45 · Professional Practice and Legal Communication

What standard must a substitute decision-maker use when making a decision for an incapable patient?

Answer

They must make the decision the incapable patient would have made if capable, known as substituted judgment, or if that is unknown, must act in the incapable patient's best interests.

#46 · Professional Practice and Legal Communication

Is capacity a fixed, all-or-nothing state?

Answer

No. Capacity is decision-specific and can fluctuate; a patient may be capable of consenting to transport while being incapable of consenting to a more complex treatment decision, and capacity should be reassessed if condition changes.

#47 · Professional Practice and Legal Communication

What four elements must be proven to establish negligence in a legal sense?

Answer

Negligence requires a duty of care owed to the patient, a breach of the standard of care expected of a reasonably prudent paramedic, damages suffered by the patient, and a causal link between the breach and the damages.

#48 · Professional Practice and Legal Communication

Once a paramedic begins patient care, can that duty of care simply be abandoned?

Answer

No. Once a duty of care exists it cannot simply be abandoned without properly transferring care to an equal or higher level of provider.

#49 · Professional Practice and Legal Communication

What defines a paramedic's scope of practice?

Answer

Scope of practice is defined by the medical directives issued under Base Hospital oversight and the certification level a paramedic holds.

#50 · Professional Practice and Legal Communication

Is it acceptable to perform an intervention outside one's authorized scope of practice if it helps the patient?

Answer

No. Performing an intervention outside one's authorized scope, even with good intentions and even if it helps the patient, exposes the paramedic and the service to liability and is a serious professional conduct issue.

#51 · Professional Practice and Legal Communication

What legislation establishes the legal framework for ambulance service delivery in Ontario, including certification requirements and vehicle standards?

Answer

The Ambulance Act and its regulations, including Ontario Regulation 257/00, establish this legal framework.

#52 · Professional Practice and Legal Communication

For AEMCA exam purposes, is it more important to memorize specific section numbers of O. Reg. 257/00 or to understand what it governs?

Answer

Understanding that the regulation exists and governs day-to-day operational and certification requirements is more important for exam purposes than memorizing specific section numbers.

#53 · Professional Practice and Legal Communication

Under Ontario's Mental Health Act, who can order the apprehension of a person believed to have a mental disorder and pose a risk of serious harm?

Answer

A physician, and in defined circumstances a police officer, can order the apprehension of such a person, documented through a Form 1.

#54 · Professional Practice and Legal Communication

What does a Form 1 authorize, and what does it not authorize?

Answer

A Form 1 authorizes transport and assessment of the person at a psychiatric facility against their will for a limited period, but it does not authorize treatment against the patient's will.

#55 · Professional Practice and Legal Communication

Do paramedics complete a Form 1 themselves?

Answer

No. Paramedics are not the ones who complete a Form 1, but they are frequently the ones who transport a patient under one.

#56 · Professional Practice and Legal Communication

When transporting a patient under a Form 1, what use-of-force standard must a paramedic follow?

Answer

The paramedic must use the minimum force reasonably necessary to safely manage the patient, proportionate to the actual risk presented, and must document it thoroughly, including what alternatives were attempted first.

#57 · Professional Practice and Legal Communication

What is the purpose of the Coroners Act in the context of prehospital care?

Answer

It requires certain deaths to be reported to a coroner, and paramedics play a role in recognizing these situations and preserving the scene when a death appears to fall under coroner jurisdiction.

#58 · Professional Practice and Legal Communication

Give three categories of death that must be reported to a coroner under the Coroners Act.

Answer

Sudden and unexpected deaths, deaths occurring during or shortly after a medical or surgical procedure, and deaths that may have resulted from violence, negligence, misconduct, or malpractice must all be reported.

#59 · Professional Practice and Legal Communication

What does scene preservation mean when a death appears to fall under coroner jurisdiction?

Answer

It means not moving the body or disturbing evidence beyond what is necessary to confirm death and ensure safety, and clearly documenting the position and condition in which the patient was found.

#60 · Professional Practice and Legal Communication

What legislation is Ontario's current child welfare law, and what duty does it create for paramedics?

Answer

The Child, Youth and Family Services Act, 2017, creates a legal duty for any person, including paramedics, with reasonable grounds to suspect a child is or may be in need of protection to report that suspicion directly to a children's aid society without delay.

#61 · Professional Practice and Legal Communication

Can a paramedic delegate their duty to report suspected child abuse to someone else, such as a nurse at the receiving hospital?

Answer

No. This duty cannot be delegated to someone else to report on the paramedic's behalf, even when the information was obtained in a professional or confidential context.

#62 · Professional Practice and Legal Communication

What threshold triggers the duty to report suspected child abuse under the Child, Youth and Family Services Act?

Answer

Reasonable grounds to suspect triggers the duty, which is a lower threshold than certainty; a paramedic who waits for proof before reporting has failed the standard.

#63 · Professional Practice and Legal Communication

What exemptions does the Highway Traffic Act grant to ambulances responding to an emergency?

Answer

It grants exemptions such as proceeding through a red signal or exceeding the posted speed limit, but only when it is safe to do so and generally requiring the use of emergency lights and siren together.

#64 · Professional Practice and Legal Communication

Do Highway Traffic Act exemptions remove a paramedic driver's responsibility for safety?

Answer

No. The exemptions do not remove a driver's responsibility to exercise due regard for the safety of others.

#65 · Professional Practice and Legal Communication

What does 'due regard' mean for a paramedic driver, and what is the legal consequence of not exercising it?

Answer

Due regard means driving in a way that a reasonable, careful driver would use given the emergency circumstances, and a driver who causes a collision while responding can still be found liable if they did not exercise due regard.

#66 · Professional Practice and Legal Communication

Give two signs on scene that may suggest abuse is occurring in a home.

Answer

Injuries inconsistent with the stated mechanism and delayed treatment-seeking are both signs, along with a patient who appears frightened of another person present.

#67 · Professional Practice and Legal Communication

How does the obligation to report abuse of a capable adult who does not wish it reported differ from reporting suspected child abuse?

Answer

Reporting abuse of a capable adult is more limited since a capable adult retains autonomy over that decision, unlike suspected child abuse, which is mandatory to report under the Child, Youth and Family Services Act.

#68 · Professional Practice and Legal Communication

If a capable adult does not want their abuse reported, what should a paramedic still do?

Answer

The paramedic should still document findings carefully and connect the patient with available resources, even though the report itself is not mandatory.

#69 · Professional Practice and Legal Communication

What standards set out what must be recorded on every ambulance call in Ontario?

Answer

Ontario Ambulance Documentation Standards set out what must be recorded, including times, findings, interventions, and the patient's response.

#70 · Professional Practice and Legal Communication

What is the Transfer of Care Report and why must it be left with the receiving provider?

Answer

It is the document handed to the receiving facility summarizing the call, and it must be completed and left with the receiving provider because it becomes part of the patient's permanent medical record.

#71 · Professional Practice and Legal Communication

What standard should documentation meet regarding speculation about diagnosis?

Answer

Documentation should be objective, factual, and free of speculation about diagnosis unless within the paramedic's scope to state.

#72 · Professional Practice and Legal Communication

Why is accurate documentation described as protecting more than just the patient?

Answer

A well-documented call protects the patient, the paramedic, and the service if the care given is ever questioned later.

#73 · Professional Practice and Legal Communication

What body mechanics principles reduce injury risk during patient lifting and transferring?

Answer

Using the legs rather than the back, keeping the load close to the body, and using appropriate equipment such as stair chairs and stretchers rather than attempting a manual carry beyond one's capability all reduce injury risk.

#74 · Professional Practice and Legal Communication

Give an example of how positioning decisions can be both clinical decisions and handling safety considerations.

Answer

Recovery position for an unconscious but breathing patient and a seated position for a patient in respiratory distress are both clinical positioning decisions that also affect handling safety.

#75 · Professional Practice and Legal Communication

What should driving manoeuvres during transport prioritize, beyond simply reaching the destination quickly?

Answer

Driving manoeuvres should prioritize patient comfort and safety in the back of the vehicle, since sudden braking, sharp turns, and aggressive acceleration can injure a patient who is not fully secured.

#76 · Professional Practice and Legal Communication

Why should a driver communicate with the attendant in the back before executing a manoeuvre that could affect patient care?

Answer

Because sudden movements of the vehicle can affect an intervention in progress or injure a patient who is not fully secured, so coordination between driver and attendant is required.

#77 · Professional Practice and Legal Communication

What defines a multiple casualty incident, or MCI?

Answer

An MCI occurs when the number of patients exceeds the resources immediately available to treat them using normal one-to-one methods.

#78 · Professional Practice and Legal Communication

How does triage change during an MCI compared to a normal single-patient call?

Answer

Triage shifts from treating the sickest patient first to sorting patients rapidly by survivability, doing the greatest good for the greatest number of patients rather than focusing on one patient at a time.

#79 · Professional Practice and Legal Communication

What four factors does an MCI triage system assess to sort patients?

Answer

It assesses ability to walk, respiration, perfusion, and mental status to sort patients rapidly by survivability.

#80 · Professional Practice and Legal Communication

List the four sorting categories used in MCI triage.

Answer

Patients are sorted into immediate, delayed, minor, and deceased or expectant categories.

#81 · Professional Practice and Legal Communication

What widely used triage system is the MCI sorting approach described in the chapter similar to?

Answer

It is similar in structure to widely used START-style triage.

#82 · Professional Practice and Legal Communication

What is the purpose of incident command at the scene of an MCI?

Answer

Incident command establishes a clear chain of authority, preventing duplicated effort, ensuring resources are directed where needed most, and giving every responder a clear reporting line.

#83 · Professional Practice and Legal Communication

Name three defined roles within an MCI incident command structure.

Answer

An incident commander, a triage officer, a treatment officer, and a transport officer are all defined roles within incident command.

#84 · Professional Practice and Legal Communication

At what point does an MCI's incident command structure become essential rather than optional?

Answer

It becomes essential once a scene grows beyond what a single crew can manage informally.

#85 · Professional Practice and Legal Communication

How should a candidate approach the professional practice and legal communication domain when studying for the AEMCA exam?

Answer

Not as a list of rules to memorize, but as the professional identity being built for the rest of a paramedic's career, since it touches every call, not just the dramatic ones.

#86 · Patient Assessment and Airway Management

What four questions does the scene survey answer before you touch the patient?

Answer

Is the scene safe, what happened, how many patients are there, and do you need more resources.

#87 · Patient Assessment and Airway Management

Why does scene safety always come before patient contact?

Answer

A paramedic who becomes a second patient helps nobody. If the scene is not safe you stage at a distance and request the resources needed to make entry safe, such as police or fire.

#88 · Patient Assessment and Airway Management

How does mechanism of injury shape your assessment before you reach the patient?

Answer

A rollover, fall from height, penetrating injury, or crush mechanism each predict specific injury patterns, raising your index of suspicion and shaping what you look for during the primary survey.

#89 · Patient Assessment and Airway Management

Why is counting patients an essential scene survey step?

Answer

An easily missed patient, such as an ejected occupant lying in a ditch, can be overlooked, and if patients exceed your resources you must shift to a multi-patient framework and call additional units.

#90 · Patient Assessment and Airway Management

What ventilation rate is used for an adult in respiratory arrest?

Answer

Roughly one breath every 6 seconds, or about 10 breaths per minute.

#91 · Patient Assessment and Airway Management

What mnemonic organizes the primary survey sequence?

Answer

LOC, A, B, C, D: level of consciousness, airway, breathing, circulation, and disability, with each life threat corrected on the spot before moving on.

#92 · Patient Assessment and Airway Management

What does the AVPU scale assess and what are its four levels?

Answer

AVPU assesses level of consciousness: alert, responsive to voice, responsive to pain, or unresponsive.

#93 · Patient Assessment and Airway Management

What does it mean clinically that a patient is alert and talking to you?

Answer

It confirms their airway is open and they are perfusing their brain adequately in that moment.

#94 · Patient Assessment and Airway Management

What airway sounds should you listen for during the primary survey?

Answer

Stridor, gurgling, snoring respirations, or complete silence in a patient who should be breathing all signal an airway problem.

#95 · Patient Assessment and Airway Management

What is the immediate correction for an unresponsive patient whose tongue is occluding the airway?

Answer

A head-tilt-chin-lift or jaw thrust, suctioning, or an airway adjunct as needed, corrected right away rather than documented for later.

#96 · Patient Assessment and Airway Management

What should you expose and examine during the breathing assessment when trauma is suspected?

Answer

Expose the chest and look for open wounds, flail segments, or asymmetrical rise and fall, and listen and feel for equal bilateral air movement.

#97 · Patient Assessment and Airway Management

Name three breathing life threats that demand immediate intervention in the primary survey.

Answer

Tension pneumothorax, flail chest, and an open chest wound, treated by sealing wounds, supporting ventilation, or preparing for advanced decompression.

#98 · Patient Assessment and Airway Management

How is major external hemorrhage managed during the circulation step?

Answer

Immediately, with direct pressure, wound packing, or a tourniquet depending on presentation, since uncontrolled hemorrhage is a rapidly fatal but preventable cause of death.

#99 · Patient Assessment and Airway Management

What does cool, pale, clammy skin suggest during the circulation assessment?

Answer

Poor perfusion and possible shock, even before a blood pressure reading is obtained.

#100 · Patient Assessment and Airway Management

What is assessed during the disability step of the primary survey?

Answer

A rapid neurological check of gross motor and sensory function in all four limbs along with pupil response, expanded further in the secondary survey.

#101 · Patient Assessment and Airway Management

What decision is made at the end of the primary survey?

Answer

A general impression and transport decision, often phrased as load and go versus stay and play, based on whether the patient is stable enough for a full scene assessment.

#102 · Patient Assessment and Airway Management

What findings push a crew toward rapid load and go transport?

Answer

Uncontrolled airway compromise, inadequate breathing, uncontrolled hemorrhage, or shock, with treatment continuing en route rather than prolonging scene time.

#103 · Patient Assessment and Airway Management

What are the two forms the secondary survey can take?

Answer

A focused exam for an isolated, clearly localized complaint, or a full head-to-toe exam for significant trauma, altered level of consciousness, or an unclear mechanism.

#104 · Patient Assessment and Airway Management

In what order does the head-to-toe exam proceed?

Answer

Scalp and skull, face, eyes, ears, nose, mouth, neck, chest, abdomen, pelvis, all four extremities, and the posterior body including the back when it is safe to log roll the patient.

#105 · Patient Assessment and Airway Management

What ventilation rate is used for a child, and how does it compare to an infant?

Answer

Roughly one breath every 2 to 3 seconds, or about 20 to 30 breaths per minute, with an infant ventilated at the higher end of that same range.

#106 · Patient Assessment and Airway Management

What does the mnemonic DCAP-BTLS stand for?

Answer

Deformities, contusions, abrasions, punctures or penetrations, burns, tenderness, lacerations, and swelling.

#107 · Patient Assessment and Airway Management

What does OPQRST explore during history taking?

Answer

The character of a symptom such as pain: onset, provocation and palliation, quality, region and radiation, severity, and time.

#108 · Patient Assessment and Airway Management

What does SAMPLE gather during history taking?

Answer

Signs and symptoms, allergies, medications, pertinent past medical history, last oral intake, and events leading up to the call.

#109 · Patient Assessment and Airway Management

How should a paramedic handle history from an unreliable historian?

Answer

Lean more heavily on physical findings, bystander accounts, medication bottles, and medical alert jewellery, and document corroborating information as coming from that source rather than the patient.

#110 · Patient Assessment and Airway Management

What device delivers a low concentration of supplemental oxygen for a mildly hypoxic patient who needs only modest support?

Answer

A nasal cannula, which is a lower flow, more comfortable device suited to modest oxygen needs.

#111 · Patient Assessment and Airway Management

What does the pediatric assessment triangle evaluate, and when is it used?

Answer

Appearance, work of breathing, and circulation to skin, assessed from across the room before hands-on contact to give a fast general impression.

#112 · Patient Assessment and Airway Management

Why can geriatric patients present atypically for serious conditions?

Answer

A serious problem such as a myocardial infarction or infection may present with vague symptoms like confusion or generalized weakness rather than the textbook presentation, complicated by sensory impairment, polypharmacy, and multiple chronic conditions.

#113 · Patient Assessment and Airway Management

List four signs of adequate ventilation with a bag-valve-mask.

Answer

Visible symmetrical chest rise and fall with each breath, improving or maintained SpO₂, improving skin colour, and an appropriate heart rate response.

#114 · Patient Assessment and Airway Management

What device delivers a high concentration of oxygen to a spontaneously breathing patient who is significantly hypoxic or in shock?

Answer

A non-rebreather mask, which delivers high concentration oxygen through a reservoir bag that must remain inflated.

#115 · Patient Assessment and Airway Management

What is the roughly normal adult blood pressure range, and why does a single reading matter less than a trend?

Answer

Roughly 90 to 140 mmHg systolic over 60 to 90 mmHg diastolic; a single reading is only a data point, while a trend across multiple readings is real clinical information.

#116 · Patient Assessment and Airway Management

What is the roughly normal adult resting respiratory rate, and why does quality matter as much as rate?

Answer

Roughly 12 to 20 breaths per minute; shallow, laboured, or irregular breathing can be dangerous even at a numerically normal rate.

#117 · Patient Assessment and Airway Management

What is the generally normal SpO2 range on room air, and name two situations where the reading can be inaccurate.

Answer

Generally 94 to 100 percent on room air; it can be inaccurate in poor perfusion states, with nail polish, in carbon monoxide poisoning, and with excessive patient movement.

#118 · Patient Assessment and Airway Management

Why is blood glucose considered a critical vital sign for altered level of consciousness?

Answer

Hypoglycemia is a rapidly reversible cause of unresponsiveness that can mimic stroke, seizure, or intoxication.

#119 · Patient Assessment and Airway Management

What is the roughly normal fasting blood glucose range in Canadian units, and the general threshold for hypoglycemia?

Answer

Roughly 4 to 7 mmol per litre normal fasting, with hypoglycemia generally defined below about 4 mmol per litre, though exact thresholds follow local protocol.

#120 · Patient Assessment and Airway Management

What three categories make up the Glasgow Coma Scale, and what is its scoring range?

Answer

Eye opening, verbal response, and motor response, summing from 3, indicating no response in any category, to 15, indicating a fully alert and oriented patient.

#121 · Patient Assessment and Airway Management

What does cyanosis indicate, and where is it most visible?

Answer

It indicates significant hypoxia and demands immediate airway and oxygenation attention, most visible around the lips and nail beds.

#122 · Patient Assessment and Airway Management

Why is excessive bag-valve-mask ventilation rate or volume dangerous?

Answer

It can cause gastric insufflation, increase the risk of vomiting and aspiration, and in a patient with a pulse can reduce venous return and cardiac output, so controlled, age-appropriate rates and volumes matter as much as ventilating at all.

#123 · Patient Assessment and Airway Management

What is the difference between the head-tilt-chin-lift and jaw thrust maneuvers?

Answer

The head-tilt-chin-lift is used for a medical patient with no suspected spinal injury, tilting the forehead back while lifting the chin, while the jaw thrust displaces the mandible forward without extending the neck and is used when spinal injury is suspected.

#124 · Patient Assessment and Airway Management

Under what condition is an oropharyngeal airway contraindicated, and why?

Answer

In a patient with an intact gag reflex, since insertion will cause vomiting and possible aspiration.

#125 · Patient Assessment and Airway Management

How is an OPA sized and inserted differently in an adult versus a pediatric patient?

Answer

It is sized from the corner of the mouth to the earlobe or angle of the jaw; in an adult it is inserted upside down and rotated 180 degrees, while in a pediatric patient it is inserted right side up using a tongue depressor to avoid soft tissue trauma.

#126 · Patient Assessment and Airway Management

How is a nasopharyngeal airway sized and inserted?

Answer

Sized from the tip of the nose to the earlobe, lubricated, and gently inserted along the floor of the nasal passage with the bevel oriented toward the septum.

#127 · Patient Assessment and Airway Management

When should an NPA be used with caution, and why?

Answer

With suspected significant facial or basilar skull fracture, due to a theoretical risk of inadvertent intracranial placement, though this risk is now understood to be lower than once taught and local protocol should guide the decision.

#128 · Patient Assessment and Airway Management

How should a responsive patient with a partial airway obstruction who is coughing forcefully be managed?

Answer

Encourage continued coughing, since a strong cough clears an obstruction more effectively than any manual manoeuvre.

#129 · Patient Assessment and Airway Management

How does management of a complete airway obstruction differ between an adult, an infant under one year, and a pregnant or significantly obese patient?

Answer

An adult or older child receives abdominal thrusts, an infant under one year receives back blows combined with chest thrusts, and a pregnant or significantly obese patient receives chest thrusts instead of abdominal thrusts.

#130 · Patient Assessment and Airway Management

What should be done if a responsive choking patient with a complete obstruction becomes unresponsive?

Answer

Begin chest compressions, checking the mouth for a visible object before each set of rescue breaths and removing it only if visible and safely retrievable, never with a blind finger sweep.

#131 · Patient Assessment and Airway Management

What is a supraglottic airway device, and when is it used?

Answer

A device that sits above the vocal cords rather than being inserted through them, used as an alternative to more invasive airway management when BVM ventilation alone is inadequate or a definitive airway cannot be achieved.

#132 · Patient Assessment and Airway Management

When is a rigid suction catheter preferred over a flexible one?

Answer

A rigid tonsil-tip catheter is preferred for the oropharynx because its larger bore clears thick material such as vomitus quickly, while a flexible catheter suits the nasopharynx or suctioning through an airway adjunct.

#133 · Patient Assessment and Airway Management

How long should a single adult suctioning attempt generally last, and when is suction applied?

Answer

About 10 to 15 seconds at a time, applied only while withdrawing the catheter, never on insertion, followed by reoxygenation before another attempt if needed.

#134 · Patient Assessment and Airway Management

Name four indications for administering supplemental oxygen in the field.

Answer

Signs of hypoxia such as a low SpO2 reading, cyanosis, altered level of consciousness with a respiratory or perfusion cause, respiratory distress, cardiac chest pain, shock, or significant trauma.

#135 · Patient Assessment and Airway Management

What is the most reliable BVM technique for achieving a good mask seal, and what should be watched to judge ventilation adequacy?

Answer

A two-person technique where one provider maintains the seal with both hands using a C-E grip while the other squeezes the bag, watching for visible chest rise rather than judging adequacy by feel alone.

#136 · Pharmacology and IV Therapy

What are the four phases of pharmacokinetics?

Answer

Absorption, distribution, metabolism, and excretion, together describing how a drug moves through and is eventually cleared from the body.

#137 · Pharmacology and IV Therapy

Why does an IV medication have the fastest, most predictable onset of any route?

Answer

It bypasses absorption entirely because it is injected directly into the bloodstream, so the full dose is available to act immediately.

#138 · Pharmacology and IV Therapy

Why do sublingual medications like nitroglycerin act quickly?

Answer

The tissue under the tongue is thin and highly vascular and it bypasses first-pass metabolism in the liver, so the drug reaches the bloodstream rapidly and at full strength.

#139 · Pharmacology and IV Therapy

A patient in shock has poorly perfused tissue. How does this affect drug distribution?

Answer

Poorly perfused tissue receives medication more slowly and unpredictably, which is one reason hypoperfused patients can respond unexpectedly to a standard dose.

#140 · Pharmacology and IV Therapy

What organ does most drug metabolism occur in, and what happens if it is diseased?

Answer

The liver metabolizes most drugs into usually less active metabolites; significant liver disease can slow this process and prolong the drug's effect.

#141 · Pharmacology and IV Therapy

What is the main route of drug excretion, and why does renal impairment matter?

Answer

Most drugs and their metabolites are excreted through the kidneys and urine; renal impairment can cause a drug to accumulate to dangerous levels in the body.

#142 · Pharmacology and IV Therapy

Distinguish an agonist from an antagonist and give the AEMCA example of an antagonist.

Answer

An agonist binds a receptor and produces a response, while an antagonist binds a receptor and blocks the normal response. Naloxone is the classic antagonist example: it binds opioid receptors without activating them, displacing opioids and reversing respiratory depression.

#143 • Pharmacology and IV Therapy

Which route is used for oral glucose in an alert hypoglycemic patient, and why is it chosen for that indication?

Answer

The oral (PO) route, since the patient is alert and can safely swallow and protect their airway; it is the slowest common route but works well when speed is less critical than airway safety.

#144 • Pharmacology and IV Therapy

What device delivers naloxone by the intranasal route, and why is this route commonly favoured?

Answer

A mucosal atomization device (MAD) sprays the drug into the highly vascular nasal mucosa; it is favoured because it requires no needle and still provides a fast onset.

#145 • Pharmacology and IV Therapy

List the standard rights of medication administration a PCP confirms before giving any drug.

Answer

Right patient, right medication, right dose, right route, right time, and right documentation, with some programs also including right indication and the patient's right to refuse.

#146 • Pharmacology and IV Therapy

Why is nitroglycerin withheld from a patient who has taken sildenafil or tadalafil within the past 24 to 48 hours?

Answer

Combining nitroglycerin with a phosphodiesterase inhibitor can cause profound, refractory hypotension, since both drugs lower blood pressure through related vascular pathways.

#147 · Pharmacology and IV Therapy

Name a common side effect each of salbutamol and diphenhydramine.

Answer

Salbutamol commonly causes tachycardia and tremor from its beta-agonist action, while diphenhydramine commonly causes drowsiness because it crosses into the central nervous system.

#148 · Pharmacology and IV Therapy

Roughly what percentage of inspired oxygen does a non-rebreather mask deliver with the reservoir bag inflated?

Answer

Close to 90 to 100 percent inspired oxygen, making it a high-flow delivery device compared with a nasal cannula's 24 to 44 percent.

#149 · Pharmacology and IV Therapy

What is the adult IM epinephrine dose for anaphylaxis, and what concentration is used?

Answer

0.3 to 0.5 mg IM using the 1:1000 concentration (1 mg/mL), injected into the anterolateral thigh and repeated after 5 to 15 minutes if the patient does not improve.

#150 · Pharmacology and IV Therapy

A 20 kg child in anaphylaxis needs weight-based epinephrine 1:1000 IM at 0.01 mg/kg. What volume is given?

Answer

0.01 mg/kg times 20 kg equals a 0.2 mg dose; at 1 mg/mL that is 0.2 mL, well under the 0.5 mg maximum single pediatric dose.

#151 · Pharmacology and IV Therapy

What is infiltration, and how does it present?

Answer

Infiltration occurs when the catheter slips out of the vein and fluid or medication is delivered into surrounding tissue instead of the bloodstream, producing swelling, coolness, and discomfort without the expected blood return on aspiration; the infusion should be stopped and the catheter removed.

#152 · Pharmacology and IV Therapy

What is the standard adult cardiac arrest dose of epinephrine, and what concentration is used?

Answer

1 mg IV or IO using the 1:10,000 concentration (0.1 mg/mL), repeated every 3 to 5 minutes per the ALS PCS.

#153 • Pharmacology and IV Therapy

Using 1:10,000 epinephrine at 0.1 mg/mL, what volume delivers the standard 1 mg adult cardiac arrest dose?

Answer

1 mg divided by 0.1 mg/mL equals 10 mL, ten times the volume that the more concentrated 1:1000 anaphylaxis solution would require for the same mass of drug.

#154 • Pharmacology and IV Therapy

What is the standard adult unit dose of nebulized salbutamol, and what condition is it used for?

Answer

5 mg mixed with normal saline and delivered by nebulizer mask, used for bronchospasm from asthma, COPD exacerbation, or other reactive airway presentations.

#155 · Pharmacology and IV Therapy

What is the standard adult dose of ASA for suspected acute coronary syndrome, and why is it chewed rather than swallowed whole?

Answer

160 to 162 mg, typically two 81 mg chewable tablets; chewing lets the drug reach the bloodstream faster through the buccal mucosa and stomach rather than passing through slower GI absorption.

#156 · Pharmacology and IV Therapy

How does ASA help in suspected acute coronary syndrome, and what are its key contraindications?

Answer

It works as an antiplatelet agent, reducing platelet clumping and limiting extension of a coronary clot; it is withheld with known ASA or NSAID allergy, active gastrointestinal bleeding, or a bleeding disorder.

#157 · Pharmacology and IV Therapy

What is the standard adult sublingual nitroglycerin dose, and how does it improve chest discomfort from cardiac ischemia?

Answer

0.4 mg per spray or tablet, repeated every 5 minutes up to the protocol maximum if blood pressure stays adequate; it dilates venous and coronary vessels, lowering cardiac preload and improving coronary blood flow.

#158 · Pharmacology and IV Therapy

How does glucagon raise blood sugar, and when is it chosen over oral glucose?

Answer

It stimulates the liver to break down stored glycogen into glucose over several minutes; it is chosen for a hypoglycemic patient who cannot safely swallow and has no IV access, since anything given orally to such a patient risks aspiration.

#159 · Pharmacology and IV Therapy

A pediatric protocol calls for a 0.5 mg glucagon dose from a vial reconstituted to 1 mg per 1 mL. What volume is drawn up?

Answer

0.5 mg divided by 1 mg/mL equals 0.5 mL.

#160 · Pharmacology and IV Therapy

What is a common adult starting dose range for naloxone, and by what routes can it be given?

Answer

0.4 to 0.8 mg, repeated every few minutes if respiratory rate and depth do not improve, given intranasally, intramuscularly, or intravenously.

#161 • Pharmacology and IV Therapy

A patient needs an 0.8 mg intranasal naloxone dose from a vial concentrated at 0.4 mg/mL. What volume is administered?

Answer

0.8 mg divided by 0.4 mg/mL equals 2 mL.

#162 • Pharmacology and IV Therapy

Why should naloxone be titrated to restore adequate breathing rather than full alertness whenever possible?

Answer

Because naloxone reverses opioid sedation abruptly, patients can wake agitated or combative, so dosing to just restore adequate ventilation avoids an unnecessarily jarring reversal while still treating the danger.

#163 · Pharmacology and IV Therapy

Why is diphenhydramine never a substitute for epinephrine in true anaphylaxis?

Answer

Diphenhydramine treats histamine-driven itching and hives but does not reverse airway swelling or cardiovascular collapse, effects only epinephrine addresses; it is used only as an adjunct alongside epinephrine.

#164 · Pharmacology and IV Therapy

What is dimenhydrinate used for, and how does it differ from ondansetron in its side effect profile?

Answer

Dimenhydrinate is an antihistamine and antiemetic used for nausea and vomiting, particularly motion-related or vertigo-associated nausea, and like diphenhydramine can cause drowsiness; ondansetron is a serotonin antagonist antiemetic that treats nausea without that sedating effect.

#165 · Pharmacology and IV Therapy

Which peripheral veins are preferred for IV cannulation, and why is the antecubital fossa less ideal despite its large veins?

Answer

The dorsal veins of the hand, the cephalic and basilic veins of the forearm, and the antecubital fossa veins at the elbow crease; the antecubital fossa offers easy access but movement at the elbow can dislodge the catheter.

#166 · Pharmacology and IV Therapy

What is phlebitis, and what does it look like on exam?

Answer

Phlebitis is inflammation of the vein, caused by mechanical or chemical irritation or infection, presenting as redness, warmth, tenderness, and a palpable cord along the affected vein.

#167 · Pharmacology and IV Therapy

At what angle and bevel orientation is an IV catheter-over-needle typically inserted?

Answer

At a shallow angle, commonly 10 to 30 degrees, bevel up, until a flashback confirms venous entry, after which the catheter is advanced while the needle is withdrawn.

#168 · Pharmacology and IV Therapy

What aseptic technique steps must a paramedic follow for every IV attempt?

Answer

Wear gloves, clean the site and let it dry before puncture, never touch the cleaned site with an ungloved or contaminated finger, and dispose of the needle immediately into an approved sharps container without recapping it.

#169 · Pharmacology and IV Therapy

How does an air embolism occur during IV therapy, and what does it cause?

Answer

Air enters the venous system, most often through poorly primed tubing or a loose connection, and can travel to the heart and lungs, causing sudden dyspnea, chest pain, and cardiovascular compromise; careful priming and monitoring of connections prevents it.

#170 · Pharmacology and IV Therapy

A patient needs 250 mL of normal saline infused over 60 minutes with tubing at a drop factor of 10 gtt/mL. What is the flow rate in gtt/min?

Answer

250 mL times 10 gtt/mL divided by 60 minutes equals approximately 41.7, rounded to 42 gtt/min.

#171 · Cardiovascular System and ECG

Where is the heart located within the thorax?

Answer

The heart is a four chamber muscular pump located in the mediastinum, slightly left of the sternal midline.

#172 · Cardiovascular System and ECG

Why is the left ventricle the most muscular chamber of the heart?

Answer

The left ventricle must generate enough pressure to push blood through the entire systemic circulation, so its wall is thicker and more muscular than the other three chambers.

#173 · Cardiovascular System and ECG

Name the two atrioventricular valves and where each sits.

Answer

The tricuspid valve sits between the right atrium and right ventricle, and the mitral (bicuspid) valve sits between the left atrium and left ventricle.

#174 · Cardiovascular System and ECG

Name the two semilunar valves and where each sits.

Answer

The pulmonic valve sits at the exit of the right ventricle, and the aortic valve sits at the exit of the left ventricle.

#175 · Cardiovascular System and ECG

Trace the path of deoxygenated blood from the right atrium to the pulmonary circulation.

Answer

Deoxygenated blood enters the right atrium, passes through the tricuspid valve into the right ventricle, and is pumped through the pulmonic valve toward the lungs for gas exchange.

#176 · Cardiovascular System and ECG

Trace the path of oxygenated blood from the pulmonary veins to the aorta.

Answer

Oxygenated blood returns through the pulmonary veins into the left atrium, passes through the mitral valve into the left ventricle, and is pumped through the aortic valve into the aorta for distribution to the body.

#177 · Cardiovascular System and ECG

What is the heart's primary pacemaker and what is its intrinsic rate?

Answer

The sinoatrial (SA) node, located in the upper right atrium, is the heart's primary pacemaker, with an intrinsic rate of about 60 to 100 beats per minute.

#178 · Cardiovascular System and ECG

What role does the AV node play in the conduction system?

Answer

The AV node briefly delays the impulse from the SA node so the atria finish emptying into the ventricles before ventricular contraction begins, and it serves as a backup pacemaker at about 40 to 60 beats per minute if the SA node fails.

#179 · Cardiovascular System and ECG

Describe the pathway the electrical impulse follows after leaving the AV node.

Answer

The impulse travels down the bundle of His, splits into the right and left bundle branches, and spreads through the Purkinje fibers, triggering ventricular contraction.

#180 · Cardiovascular System and ECG

What is the intrinsic backup rate of the Purkinje system, and what is this rhythm sometimes called?

Answer

The Purkinje system has the slowest backup rate, about 20 to 40 beats per minute, sometimes called an idioventricular rhythm.

#181 · Cardiovascular System and ECG

Which areas of the heart does the right coronary artery typically supply?

Answer

The right coronary artery (RCA) typically supplies the right atrium, right ventricle, and in most people the inferior wall of the left ventricle and the AV node.

#182 · Cardiovascular System and ECG

What territories do the LAD and left circumflex arteries supply?

Answer

The left anterior descending (LAD) artery supplies the anterior wall and much of the septum, while the left circumflex artery supplies the lateral wall of the left ventricle.

#183 · Cardiovascular System and ECG

Define stroke volume and cardiac output, with typical values.

Answer

Stroke volume is the blood ejected by the left ventricle per contraction, typically 60 to 100 millilitres at rest. Cardiac output is stroke volume multiplied by heart rate, normally roughly 4 to 6 litres per minute.

#184 · Cardiovascular System and ECG

Distinguish preload, afterload, and contractility.

Answer

Preload is the volume stretching the ventricle at the end of diastole. Afterload is the resistance the ventricle must pump against, largely from systemic vascular resistance and arterial pressure. Contractility is the intrinsic strength of contraction, independent of preload or afterload.

#185 · Cardiovascular System and ECG

What is atherosclerosis, and what is the most common mechanism behind an acute myocardial infarction?

Answer

Atherosclerosis is the progressive buildup of fatty plaque within artery walls that narrows and stiffens vessels over years. When a coronary plaque ruptures it can trigger a clot that suddenly occludes the vessel, the most common mechanism behind acute myocardial infarction.

#186 · Cardiovascular System and ECG

List the major risk factors for atherosclerosis.

Answer

Risk factors include smoking, hypertension, diabetes, high cholesterol, obesity, sedentary lifestyle, and family history.

#187 · Cardiovascular System and ECG

Distinguish ischemia from infarction.

Answer

Ischemia is a temporary reduction in blood flow and oxygen delivery that causes pain and dysfunction but does not necessarily cause permanent cell death if flow is restored quickly. Infarction is actual tissue death from prolonged, complete loss of flow.

#188 · Cardiovascular System and ECG

What happens to cardiac muscle cells after they infarct, and why does this make time critical?

Answer

Infarcted cardiac muscle cells do not regenerate, they are replaced by scar tissue, which is why rapid transport for suspected myocardial infarction is one of the most time critical interventions a paramedic performs, captured by the phrase "time is muscle."

#189 · Cardiovascular System and ECG

What causes the signs and symptoms of left sided heart failure?

Answer

Left sided heart failure causes blood to back up into the pulmonary circulation, leading to pulmonary edema, shortness of breath, orthopnea, crackles on lung auscultation, and often anxiety from air hunger.

#190 · Cardiovascular System and ECG

What causes the signs and symptoms of right sided heart failure?

Answer

Right sided heart failure causes blood to back up into the systemic venous circulation, leading to peripheral edema, jugular venous distension, and hepatic congestion.

#191 · Cardiovascular System and ECG

How does left sided heart failure typically lead to right sided heart failure over time?

Answer

A failing left ventricle raises pressure that backs up through the lungs and strains the right side of the heart, so left sided failure is the more common cause of right sided failure over time.

#192 · Cardiovascular System and ECG

What distinguishes a true hypertensive emergency from simply having a high blood pressure reading?

Answer

A hypertensive emergency involves blood pressure high enough to cause or threaten acute organ damage to the brain, heart, kidneys, or eyes, unlike an asymptomatic high reading with no symptoms.

#193 · Cardiovascular System and ECG

What signs suggest a true hypertensive emergency?

Answer

Signs can include severe headache, altered mental status, visual disturbance, chest pain, or signs of stroke, and these patients need prompt transport and monitoring rather than aggressive prehospital blood pressure lowering.

#194 · Cardiovascular System and ECG

What is cardiac tamponade, and what is the classic triad of findings?

Answer

Cardiac tamponade occurs when fluid, often blood, accumulates in the pericardial sac faster than it can be absorbed, compressing the heart and restricting diastolic filling. The classic Beck triad is muffled heart sounds, jugular venous distension, and hypotension, along with narrowing pulse pressure and pulsus paradoxus.

#195 · Cardiovascular System and ECG

What two general mechanisms give rise to dysrhythmias?

Answer

Dysrhythmias arise from problems with impulse formation, such as an abnormal ectopic focus firing instead of the SA node, or problems with conduction, such as a block, delay, or re-entry circuit.

#196 · Cardiovascular System and ECG

Compare 3-lead and 12-lead ECG configurations.

Answer

A 3-lead configuration uses electrodes on the right arm, left arm, and left leg for continuous rhythm monitoring during transport. A 12-lead adds six chest (precordial) leads to generate twelve views of the heart's electrical activity and is essential for localizing ischemia and recognizing STEMI.

#197 · Cardiovascular System and ECG

Explain the 6 second method for estimating heart rate from an ECG strip.

Answer

The 6 second method counts the number of QRS complexes in a 6 second strip and multiplies that count by 10; it is generally the most reliable method for irregular rhythms.

#198 · Cardiovascular System and ECG

Explain the large box and small box methods for estimating heart rate.

Answer

The large box method divides 300 by the number of large boxes between two consecutive R waves. The small box method divides 1500 by the small boxes between R waves for more precision. Both suit regular rhythms.

#199 · Cardiovascular System and ECG

List the four steps of a systematic rhythm analysis approach.

Answer

First assess regularity by comparing R wave spacing, second identify whether consistent P waves precede each QRS, third measure the PR interval, and fourth measure QRS width.

#200 · Cardiovascular System and ECG

What is a normal PR interval, and what does it represent?

Answer

A normal PR interval is 0.12 to 0.20 seconds (120 to 200 milliseconds), reflecting conduction time through the AV node.

#201 · Cardiovascular System and ECG

What is a normal QRS duration, and what does a widened QRS suggest?

Answer

A normal QRS duration is less than 0.12 seconds (120 milliseconds). A widened QRS suggests the impulse originates from or is conducted abnormally through ventricular tissue rather than the normal conduction pathway.

#202 · Cardiovascular System and ECG

Describe the ECG features of normal sinus rhythm.

Answer

Normal sinus rhythm shows a regular rate between 60 and 100 beats per minute, a normal upright P wave before every QRS, a normal PR interval, and a narrow QRS.

#203 · Cardiovascular System and ECG

How do sinus tachycardia and sinus bradycardia differ from normal sinus rhythm?

Answer

Sinus tachycardia is a sinus rhythm faster than 100 beats per minute and sinus bradycardia is a sinus rhythm slower than 60; both retain a normal P wave before every QRS and a normal PR interval, only the rate is outside range.

#204 · Cardiovascular System and ECG

Give common causes of sinus tachycardia and explain why treating the rate alone is usually wrong.

Answer

Sinus tachycardia is often compensatory, a response to pain, fever, anxiety, dehydration, blood loss, or hypoxia, so the underlying cause needs treatment rather than the rate itself.

#205 · Cardiovascular System and ECG

What determines whether sinus bradycardia needs treatment?

Answer

Sinus bradycardia may be normal in a fit athlete or may reflect increased vagal tone, hypoxia, or certain medications, and management depends on whether the patient is symptomatic, such as hypotensive, altered, or in shock, or not.

#206 · Cardiovascular System and ECG

Describe the ECG appearance of atrial fibrillation.

Answer

Atrial fibrillation appears as an irregularly irregular rhythm with no discernible, consistent P waves, often replaced by a fine or coarse fibrillatory baseline, reflecting chaotic, disorganized atrial electrical activity.

#207 · Cardiovascular System and ECG

Why can the ventricular rate in atrial fibrillation vary from slow to rapid?

Answer

The ventricular rate depends on how many of the chaotic atrial impulses conduct through the AV node, so it can be slow, normal, or rapid.

#208 · Cardiovascular System and ECG

Describe the ECG features of supraventricular tachycardia (SVT) and the mechanism that usually produces it.

Answer

SVT is a fast, regular rhythm, usually well over 150 beats per minute, with P waves frequently absent or hidden in the preceding T wave, and a typically narrow QRS. It often arises from a re-entry circuit involving the AV node.

#209 · Cardiovascular System and ECG

Describe the ECG features of ventricular tachycardia (VT).

Answer

VT is a rapid, often regular rhythm with wide, bizarre looking QRS complexes and no clearly associated P waves, originating from an irritable focus or re-entry circuit within the ventricles that bypasses the normal conduction system.

#210 · Cardiovascular System and ECG

How does management of pulseless VT differ from VT with a pulse?

Answer

Pulseless VT is treated as a shockable cardiac arrest rhythm, while VT with a pulse may be stable or unstable depending on blood pressure, level of consciousness, and signs of poor perfusion.

#211 · Cardiovascular System and ECG

Describe the ECG appearance of ventricular fibrillation (VF).

Answer

VF shows an erratic waveform with no identifiable P waves, QRS complexes, or T waves, reflecting chaotic, disorganized electrical activity throughout the ventricles with no organized contraction.

#212 · Cardiovascular System and ECG

Is VF ever seen with a pulse, and what is the immediate treatment?

Answer

There is no cardiac output in VF, it is always a pulseless, life threatening rhythm requiring immediate defibrillation.

#213 · Cardiovascular System and ECG

What is asystole, and what must be checked before confirming it?

Answer

Asystole is the complete absence of detectable electrical activity, appearing as a flat or nearly flat line; before confirming it, paramedics must check lead connections, gain, and a second lead view, since fine VF can resemble a flat line in one lead.

#214 · Cardiovascular System and ECG

Define pulseless electrical activity (PEA).

Answer

PEA describes any organized rhythm on the monitor that would normally be expected to produce a pulse, occurring in a patient with no palpable pulse; it is not a specific waveform but a clinical situation where electrical and mechanical activity have become disconnected, often from a reversible cause.

#215 · Cardiovascular System and ECG

Describe the ECG finding in first degree AV block.

Answer

First degree AV block shows a prolonged but constant PR interval greater than 0.20 seconds, with every P wave still followed by a QRS.

#216 · Cardiovascular System and ECG

Describe the two patterns of second degree AV block.

Answer

One pattern shows the PR interval progressively lengthening until a QRS drops entirely, and the other shows a constant PR interval but with QRS complexes intermittently dropped.

#217 · Cardiovascular System and ECG

What defines third degree (complete) AV block?

Answer

In third degree AV block, no atrial impulses reach the ventricles at all, so the atria and ventricles beat independently, each at their own intrinsic rate.

#218 · Cardiovascular System and ECG

A patient's strip shows a regular narrow QRS rhythm at 88 beats per minute with a normal upright P wave before every QRS and a PR interval of 0.16 seconds. What rhythm is this?

Answer

This is normal sinus rhythm: the rate falls within 60 to 100, every QRS is preceded by a normal P wave, the PR interval is within the normal 0.12 to 0.20 second range, and the QRS is narrow.

#219 · Cardiovascular System and ECG

A strip shows a rapid, regular, wide QRS rhythm with no clear P waves in a patient who is pulseless. What rhythm is this and how is it treated?

Answer

This is pulseless ventricular tachycardia, and it is treated as a shockable cardiac arrest rhythm with defibrillation.

#220 · Cardiovascular System and ECG

What rhythms is defibrillation indicated for, and why are asystole and PEA excluded?

Answer

Defibrillation is indicated only for shockable rhythms, ventricular fibrillation and pulseless ventricular tachycardia. Asystole and PEA are never shocked because there is no chaotic electrical activity to interrupt, so management instead focuses on high quality CPR and identifying reversible causes.

#221 · Cardiovascular System and ECG

What happens to the heart's electrical activity during a defibrillation shock?

Answer

Defibrillation delivers an unsynchronized electrical shock through the heart, momentarily stopping all electrical activity at once and allowing the heart's natural pacemaker, usually the SA node, to resume organized activity afterward.

#222 · Cardiovascular System and ECG

How does an AED differ from a manual defibrillator?

Answer

An AED analyzes the rhythm itself and advises whether a shock is indicated using voice and visual prompts, designed for both trained providers and lay rescuers, while a manual defibrillator lets the paramedic view the rhythm directly and make the clinical decision to shock.

#223 · Cardiovascular System and ECG

What safety checks must be completed before delivering a defibrillation shock?

Answer

The rescuer must ensure no one is touching the patient, verbally and visually clear the area, avoid contact with excess water or metal, and check that supplemental oxygen is not aimed across the chest during the shock.

#224 · Cardiovascular System and ECG

Describe the two acceptable defibrillation pad placement positions.

Answer

Pads can be placed in the anterolateral position, upper right chest below the clavicle and lower left mid axillary line, or the anteroposterior position, anterior chest and posterior back; both pass current through the heart's mass.

#225 · Cardiovascular System and ECG

How does synchronized cardioversion differ from standard defibrillation, and when is it used?

Answer

Synchronized cardioversion times the shock to the R wave, avoiding the vulnerable T wave when a poorly timed shock could trigger VF. It is used for unstable tachyarrhythmias with a pulse, such as unstable SVT, unstable atrial fibrillation with rapid ventricular response, or unstable VT with a pulse.

#226 · Cardiovascular System and ECG

Why can unsynchronized defibrillation never be used for a rhythm with a pulse the way synchronized cardioversion can?

Answer

Unsynchronized defibrillation is reserved for pulseless, chaotic rhythms like VF and pulseless VT, where there is no organized QRS to synchronize to at all.

#227 · Cardiovascular System and ECG

What history-gathering approach is used for suspected cardiac chest pain?

Answer

Paramedics use an approach such as OPQRST, covering onset, provocation, quality, radiation, severity, and timing, alongside relevant past medical history, medications, and allergies.

#228 · Cardiovascular System and ECG

How is classic cardiac chest pain typically described, and who may present atypically?

Answer

Classic cardiac chest pain is often described as pressure, tightness, or heaviness rather than sharp pain, and may radiate to the jaw, neck, shoulder, or arm; women, older adults, and people with diabetes may present atypically, with fatigue, nausea, or shortness of breath and little chest discomfort.

#229 · Cardiovascular System and ECG

How should ASA be administered under the suspected cardiac ischemia protocol, and why?

Answer

Acetylsalicylic acid (ASA) is chewed rather than swallowed whole for faster absorption, provided there is no contraindication such as a true allergy or active bleeding concern.

#230 · Cardiovascular System and ECG

How should supplemental oxygen be applied during a suspected cardiac event?

Answer

Oxygen is titrated to maintain adequate saturation rather than applied routinely, since excess supplemental oxygen in an already well saturated patient offers no benefit and may be harmful.

#231 · Cardiovascular System and ECG

What must be checked before giving nitroglycerin, and when is it withheld?

Answer

Blood pressure must be checked first; nitroglycerin is withheld in patients who are hypotensive or who have recently used erectile dysfunction medications, due to the risk of a severe drop in blood pressure.

#232 · Cardiovascular System and ECG

Why is acquiring a 12-lead ECG early such a priority in suspected cardiac ischemia?

Answer

ST segment elevation consistent with STEMI lets the receiving hospital prepare for rapid intervention, and many systems allow prehospital transmission ahead of arrival, supporting rapid transport since delays in restoring blood flow worsen outcomes.

#233 · Cardiovascular System and ECG

What is the foundation of cardiac arrest management, and what does high quality CPR require?

Answer

High quality CPR is the foundation of cardiac arrest management, since chest compressions maintain any blood flow to the brain and heart during arrest; it requires an adequate rate and depth, allowing full chest recoil, and minimizing interruptions.

#234 · Cardiovascular System and ECG

What is the standard adult compression to ventilation ratio without an advanced airway, and how does this change once one is placed?

Answer

The standard adult ratio without an advanced airway is 30 compressions to 2 ventilations. Once an advanced airway, such as a supraglottic device or endotracheal tube, is in place, compressions become continuous without pausing for breaths, with ventilations delivered at a steady rate independent of the compression cycle.

#235 · Cardiovascular System and ECG

Why should interruptions to chest compressions be kept as brief as possible?

Answer

Even short pauses, such as for rhythm checks or shock delivery, reduce perfusion pressure that takes time to rebuild, so interruptions should be minimized.

#236 · Cardiovascular System and ECG

Why is epinephrine given during cardiac arrest, and how often is it typically repeated?

Answer

Epinephrine increases coronary and cerebral perfusion pressure through its vasoconstrictive effect, typically given intravenously or intraosseously and repeated roughly every 3 to 5 minutes, per current medical direction.

#237 · Cardiovascular System and ECG

List the Hs and Ts of reversible cardiac arrest causes.

Answer

The Hs are hypovolemia, hypoxia, hydrogen ion excess (acidosis), hypo or hyperkalemia, and hypothermia. The Ts are tension pneumothorax, cardiac tamponade, toxins, and thrombosis, whether coronary or pulmonary.

#238 · Cardiovascular System and ECG

How does acute heart failure with pulmonary edema typically present?

Answer

It presents with significant respiratory distress, often sudden in onset, the patient preferring to sit upright, using accessory muscles, showing anxiety from air hunger, with crackles on auscultation and sometimes pink, frothy sputum.

#239 · Cardiovascular System and ECG

How is acute heart failure with pulmonary edema managed in the field?

Answer

Management focuses on positioning the patient upright with legs dependent, providing oxygen or ventilatory support up to and including CPAP where available, following protocol guidance on medications that reduce cardiac workload, and reassessing respiratory status and vital signs.

#240 · Cardiovascular System and ECG

What should prehospital management of a hypertensive emergency focus on?

Answer

Paramedics should assess for neurological changes, visual disturbance, chest pain, or shortness of breath alongside the reading, obtain a full set of vital signs, and provide supportive care and rapid transport, since aggressive prehospital blood pressure reduction outside specific protocol guidance can be dangerous.

#241 · Respiratory and Nervous Systems

What structures make up the upper airway?

Answer

The nose, mouth, pharynx (nasopharynx, oropharynx, and laryngopharynx), and larynx make up the upper airway.

#242 · Respiratory and Nervous Systems

What is the function of the epiglottis?

Answer

The epiglottis is a leaf-shaped flap of cartilage that folds over the tracheal opening during swallowing to prevent aspiration of food or liquid into the lower airway.

#243 · Respiratory and Nervous Systems

Why is the tongue the most common cause of airway obstruction in an unconscious patient?

Answer

In an unconscious patient the tongue can fall posteriorly against the pharyngeal wall, making the upper airway the most common site of obstruction.

#244 · Respiratory and Nervous Systems

Why do aspirated foreign bodies and misplaced endotracheal tubes tend to end up in the right lung?

Answer

The right mainstem bronchus is shorter, wider, and more vertically oriented than the left, so aspirated objects and misplaced tubes tend to travel into the right lung.

#245 · Respiratory and Nervous Systems

Where does gas exchange actually take place in the lungs?

Answer

Gas exchange occurs in the alveoli, the microscopic air sacs at the end of the bronchioles, across the thin alveolar-capillary membrane.

#246 · Respiratory and Nervous Systems

Describe the mechanics of inhalation.

Answer

The diaphragm contracts and flattens, the external intercostals lift the ribs, and the thoracic cavity expands, dropping intrathoracic pressure below atmospheric pressure and drawing air in, a process called negative-pressure ventilation.

#247 · Respiratory and Nervous Systems

Which muscles are recruited when the work of breathing increases, and why does their use matter clinically?

Answer

The sternocleidomastoids and scalenes are recruited as accessory muscles when breathing becomes harder, and visible use of these muscles is one of the clearest field signs of respiratory distress.

#248 · Respiratory and Nervous Systems

What two processes must be matched for effective gas exchange to occur?

Answer

Ventilation, the movement of air in and out of the alveoli, must be reasonably matched with perfusion, the blood flow through the pulmonary capillaries, a relationship often called the ventilation-perfusion match.

#249 · Respiratory and Nervous Systems

What three processes cause the airway narrowing seen in asthma?

Answer

Asthma narrows the airways through bronchospasm, mucosal edema, and increased mucus production, all of which raise airway resistance.

#250 · Respiratory and Nervous Systems

Why does asthma classically cause expiratory wheezing rather than difficulty breathing in?

Answer

Air can usually be drawn in past the narrowed airways but struggles to get back out, trapping air in the alveoli and increasing lung volume, which produces expiratory wheezing.

#251 · Respiratory and Nervous Systems

Why is a silent chest during a severe asthma attack an ominous sign rather than a sign of improvement?

Answer

In a severe attack airflow can drop so low that it no longer generates turbulent sound, so the wheeze disappears even though the patient is deteriorating, not improving.

#252 · Respiratory and Nervous Systems

What two conditions does the umbrella term COPD cover?

Answer

COPD covers chronic bronchitis and emphysema, both usually caused by long-term exposure to cigarette smoke or occupational irritants.

#253 · Respiratory and Nervous Systems

How does chronic bronchitis typically present, and what informal nickname describes it?

Answer

Chronic bronchitis causes a chronic productive cough and often a bluish, cyanotic appearance from chronic hypoxemia, sometimes informally called a blue bloater presentation.

#254 · Respiratory and Nervous Systems

What underlying process causes emphysema, and how does it present?

Answer

Emphysema involves destruction of the alveolar walls and loss of elastic recoil, trapping air in the lungs and producing a barrel-chested appearance with pursed-lip breathing, sometimes called a pink puffer presentation.

#255 · Respiratory and Nervous Systems

Should oxygen be withheld from a hypoxic COPD patient out of concern for hypoxic drive?

Answer

No. Untreated hypoxia is immediately life-threatening, so oxygen should still be given but titrated to an appropriate target saturation with close monitoring of respiratory effort.

#256 · Respiratory and Nervous Systems

How does pneumonia impair gas exchange?

Answer

Pneumonia is an infection of the lung parenchyma that fills the alveoli with inflammatory fluid, pus, and cellular debris, and this consolidation impairs gas exchange in the affected lung segment.

#257 · Respiratory and Nervous Systems

Where does the thrombus causing a pulmonary embolism most commonly originate?

Answer

A pulmonary embolism most commonly originates as a thrombus in the deep veins of the legs that travels through the venous system and lodges in the pulmonary arterial circulation.

#258 · Respiratory and Nervous Systems

List the classic findings of pulmonary embolism.

Answer

Classic findings include sudden-onset pleuritic chest pain, dyspnea, tachycardia, and in some cases hemoptysis, though presentation varies widely.

#259 · Respiratory and Nervous Systems

What is the difference between an open pneumothorax and a tension pneumothorax?

Answer

An open pneumothorax involves a chest wound that communicates directly with the pleural space, letting air move freely in and out; a tension pneumothorax involves a one-way valve effect that lets air enter but not escape, progressively building pressure.

#260 · Respiratory and Nervous Systems

What happens physiologically as a tension pneumothorax progresses if left untreated?

Answer

Pressure builds progressively, collapsing the affected lung and eventually pushing the mediastinum toward the opposite side, kinking the great vessels, reducing venous return, and rapidly progressing to obstructive shock and cardiac arrest.

#261 · Respiratory and Nervous Systems

What is the most common prehospital cause of pulmonary edema?

Answer

Acute left-sided heart failure is the most common prehospital cause, since the failing left ventricle cannot keep pace with venous return, raising pulmonary capillary pressure and forcing fluid across the alveolar-capillary membrane.

#262 · Respiratory and Nervous Systems

What are the typical presenting features of acute pulmonary edema?

Answer

Severe dyspnea, orthopnea, diffuse crackles throughout the lung fields, pink frothy sputum in severe cases, and significant anxiety from the sensation of drowning in their own fluid.

#263 · Respiratory and Nervous Systems

What does the tripod position tell you about a patient's respiratory status?

Answer

A patient who insists on sitting upright and leaning forward is signalling that their airway and breathing status is compromised.

#264 · Respiratory and Nervous Systems

How do wheezes sound and what condition are they classically associated with?

Answer

Wheezes are high-pitched musical sounds produced by air moving through narrowed airways and are classically associated with asthma and bronchospasm.

#265 · Respiratory and Nervous Systems

What causes crackles (rales) and in which conditions are they heard?

Answer

Crackles are discontinuous popping sounds caused by fluid in the alveoli or the sudden opening of collapsed small airways, and are heard in pulmonary edema and pneumonia.

#266 · Respiratory and Nervous Systems

What are rhonchi and what causes them?

Answer

Rhonchi are lower-pitched, rattling breath sounds caused by mucus or secretions in the larger airways, often clearing partially with coughing.

#267 · Respiratory and Nervous Systems

What does stridor indicate and why is it a true emergency?

Answer

Stridor is a harsh, high-pitched sound heard primarily on inspiration that indicates upper airway obstruction and represents a true airway emergency requiring immediate intervention.

#268 · Respiratory and Nervous Systems

How does salbutamol relieve bronchospasm and how is it typically delivered?

Answer

Salbutamol stimulates beta-2 adrenergic receptors in bronchial smooth muscle, relaxing and dilating the airways; it is most commonly delivered by metered-dose inhaler with a spacer or by nebulizer.

#269 · Respiratory and Nervous Systems

Why can a diminishing wheeze combined with worsening distress be a dangerous sign after bronchodilator treatment?

Answer

It can indicate progression toward a silent chest, meaning airflow is dropping too low to make sound, rather than true improvement, so work of breathing and oxygen saturation must be reassessed alongside breath sounds.

#270 · Respiratory and Nervous Systems

What is the classic constellation of findings for tension pneumothorax?

Answer

Severe respiratory distress, absent or markedly diminished breath sounds on the affected side, tracheal deviation away from the affected side in advanced cases, distended neck veins, and signs of obstructive shock such as hypotension and tachycardia.

#271 · Respiratory and Nervous Systems

Why should tracheal deviation not be relied upon as the sole criterion for diagnosing tension pneumothorax?

Answer

Tracheal deviation is often a late finding in tension pneumothorax, so waiting for it risks missing the diagnosis earlier when other signs are already present.

#272 · Respiratory and Nervous Systems

What defines a flail chest injury and what is paradoxical movement?

Answer

Flail chest occurs when three or more adjacent ribs are each fractured in two or more places, creating a free-floating segment; that segment shows paradoxical movement, drawn inward during inhalation and pushed outward during exhalation.

#273 · Respiratory and Nervous Systems

How should an occlusive dressing be applied to an open pneumothorax wound, and why?

Answer

It should be taped on three sides, allowing air to escape the pleural space on exhalation while preventing air from being drawn in on inhalation, creating a functional one-way valve that reduces the risk of the wound converting into a tension pneumothorax.

#274 · Respiratory and Nervous Systems

What is pulmonary contusion and how does its symptom course differ from a simple rib fracture?

Answer

Pulmonary contusion is bruising of the lung tissue causing localized bleeding and edema within the alveoli that impairs gas exchange; unlike a simple rib fracture, its symptoms may worsen progressively over the hours after injury rather than being maximal at the outset.

#275 · Respiratory and Nervous Systems

Why might a patient with suspected flail chest be allowed to lie on the injured side?

Answer

Lying on the injured side can help splint the flail segment and reduce paradoxical movement, though spinal precautions and mechanism of injury must still be weighed alongside patient comfort.

#276 · Respiratory and Nervous Systems

What are the two major divisions of the nervous system?

Answer

The central nervous system, comprising the brain and spinal cord, and the peripheral nervous system, comprising the cranial nerves and spinal nerves connecting the CNS to the rest of the body.

#277 · Respiratory and Nervous Systems

What are the primary functions of the cerebrum, cerebellum, and brainstem?

Answer

The cerebrum handles higher cognitive function, voluntary movement, and sensory processing; the cerebellum coordinates balance and fine motor movement; the brainstem controls vital automatic functions including respiration and heart rate.

#278 · Respiratory and Nervous Systems

How many pairs of cranial nerves arise from the brain and brainstem, and why is knowing their function useful in the field?

Answer

Twelve pairs of cranial nerves arise from the brain and brainstem, and a deficit isolated to a specific cranial nerve distribution can help localize the site of a stroke or injury.

#279 · Respiratory and Nervous Systems

How does the sympathetic nervous system change vital signs during the fight-or-flight response?

Answer

It increases heart rate, dilates pupils, redirects blood flow to skeletal muscle, and increases respiratory rate.

#280 · Respiratory and Nervous Systems

Which nerve mediates most parasympathetic rest-and-digest activity, and what effect does it have on heart rate?

Answer

The vagus nerve mediates most parasympathetic activity and slows heart rate while promoting digestion.

#281 • Respiratory and Nervous Systems

What is the difference between ischemic and hemorrhagic stroke?

Answer

Ischemic stroke results from a thrombus or embolus occluding a cerebral artery and cutting off blood supply; hemorrhagic stroke results from a ruptured blood vessel bleeding into or around brain tissue, causing damage from both lost blood flow and direct pressure.

#282 • Respiratory and Nervous Systems

Can paramedics reliably distinguish ischemic from hemorrhagic stroke on clinical grounds alone?

Answer

No, both types can present with similar focal deficits, which is why rapid transport to a facility capable of definitive imaging is critical rather than attempting to guess the stroke type on scene.

#283 · Respiratory and Nervous Systems

Describe the phases of a generalized tonic-clonic seizure.

Answer

It typically progresses through a tonic phase of sustained muscle rigidity followed by a clonic phase of rhythmic jerking movements, often followed by a postictal period of altered consciousness, confusion, and fatigue lasting minutes to hours.

#284 · Respiratory and Nervous Systems

Name several possible underlying causes of seizures besides epilepsy.

Answer

Fever in young children, hypoglycemia, head trauma, stroke, toxic ingestion, alcohol withdrawal, and eclampsia in pregnancy can all cause seizures.

#285 · Respiratory and Nervous Systems

What is Cushing's triad and what does it indicate?

Answer

Cushing's triad is rising blood pressure with a widening pulse pressure, a falling heart rate, and an irregular respiratory pattern; it indicates severe, advanced intracranial pressure elevation and often precedes herniation and death if not addressed urgently.

#286 · Respiratory and Nervous Systems

What is the difference between primary and secondary traumatic brain injury?

Answer

Primary injury occurs at the moment of impact, while secondary injury develops afterward from processes such as swelling, bleeding, hypoxia, and hypotension, all of which paramedics can influence through careful management.

#287 · Respiratory and Nervous Systems

What is coup-contrecoup injury?

Answer

Coup-contrecoup injury describes brain damage occurring both at the site of direct impact and at the opposite side of the brain as it rebounds against the skull.

#288 · Respiratory and Nervous Systems

How does syncope differ from a seizure?

Answer

Syncope is a transient loss of consciousness from a temporary reduction in cerebral blood flow, followed by spontaneous and complete recovery; it typically involves a brief loss of tone without prolonged convulsive activity and a rapid return to baseline mental status, unlike a seizure.

#289 · Respiratory and Nervous Systems

Name common causes of syncope.

Answer

Common causes include vasovagal episodes, orthostatic hypotension, cardiac dysrhythmias, and situational triggers such as coughing or straining.

#290 · Respiratory and Nervous Systems

What does the F in the FAST stroke screening framework stand for and what is assessed?

Answer

F stands for Face, assessing for asymmetric drooping.

#291 · Respiratory and Nervous Systems

What does the A in FAST assess?

Answer

A stands for Arms, assessing for unilateral drift or weakness when both arms are held out.

#292 · Respiratory and Nervous Systems

What does the S in FAST assess?

Answer

S stands for Speech, assessing for slurring or word-finding difficulty.

#293 · Respiratory and Nervous Systems

Why is the T in FAST, Time, considered one of the single most important pieces of information to gather in a stroke call?

Answer

The exact time the patient was last known to be well directly determines eligibility for time-sensitive hospital treatments.

#294 · Respiratory and Nervous Systems

What three components make up the Glasgow Coma Scale?

Answer

Eye opening, verbal response, and motor response, each scored independently and then summed to a total ranging from three to fifteen.

#295 · Respiratory and Nervous Systems

What is the range of possible Glasgow Coma Scale scores?

Answer

The GCS total ranges from three, the lowest possible score, to fifteen, indicating a fully alert and oriented patient.

#296 · Respiratory and Nervous Systems

Why is a falling GCS score often more clinically significant than the absolute number itself?

Answer

The GCS is used both to establish a baseline severity and to track trends over time, and a falling trend signals active neurological deterioration even if the current number still seems moderate.

#297 · Respiratory and Nervous Systems

What two actions should always be avoided when managing an active convulsive seizure?

Answer

Never restrain the convulsive movements and never place anything in the patient's mouth.

#298 · Respiratory and Nervous Systems

What is status epilepticus and why is it a true emergency?

Answer

Status epilepticus is a seizure lasting an extended period, or seizures occurring in rapid succession without full recovery between them, requiring prompt intervention within local protocols.

#299 · Respiratory and Nervous Systems

What does the AEIOU-TIPS mnemonic help paramedics organize?

Answer

It organizes possible causes of altered level of consciousness, including alcohol, epilepsy or endocrine disturbance, insulin-related causes, overdose or oxygen deficiency, uremia, trauma, infection, psychiatric causes, and stroke or space-occupying lesion.

#300 · Respiratory and Nervous Systems

Why does neurogenic shock present with hypotension and a normal or low heart rate instead of the tachycardia seen in hypovolemic shock?

Answer

A cervical or high thoracic spinal cord injury disrupts sympathetic nervous system output, causing widespread vasodilation and loss of vascular tone, and the sympathetic pathways that would normally drive reflex tachycardia are themselves disrupted below the injury level.

#301 · Other Body Systems

What are the two hormone-producing cell types in the islets of Langerhans, and what does each secrete?

Answer

Beta cells produce insulin, which moves glucose from the blood into cells and promotes glycogen storage. Alpha cells produce glucagon, which stimulates the liver to break down stored glycogen and release glucose into the blood.

#302 · Other Body Systems

Contrast the underlying cause of Type 1 and Type 2 diabetes.

Answer

Type 1 diabetes is an autoimmune condition where the pancreas produces little or no insulin, usually diagnosed in childhood or young adulthood and requiring lifelong insulin. Type 2 diabetes usually develops gradually in adults from insulin resistance, where cells stop responding effectively to insulin even though the pancreas may still produce it.

#303 · Other Body Systems

Why do patients in diabetic ketoacidosis develop Kussmaul respirations?

Answer

Severe insulin lack forces the body to break down fat for fuel, producing ketones that cause acidosis. Kussmaul breathing, deep and rapid respirations, is the body's attempt to blow off carbon dioxide and compensate for that acidosis.

#304 · Other Body Systems

List the classic presentation of a patient in diabetic ketoacidosis.

Answer

Gradual onset over hours to days with Kussmaul breathing, a fruity or acetone breath odour, extreme thirst, frequent urination, abdominal pain, nausea and vomiting, warm dry skin, and progressive dehydration, with mental status deteriorating as it worsens.

#305 • Other Body Systems

What field management does DKA require from a paramedic?

Answer

Supportive care only: high-flow oxygen if indicated, careful airway management given the risk of vomiting, cautious fluid administration per local protocol, and prompt transport, since definitive insulin therapy and electrolyte correction can only be delivered in hospital.

#306 • Other Body Systems

How quickly does hypoglycemia typically develop compared to hyperglycemia, and why is that dangerous?

Answer

Hypoglycemia tends to develop quickly, sometimes within minutes, and is immediately dangerous because the brain is starved of its primary fuel source, unlike hyperglycemia which usually builds gradually over hours to days.

#307 · Other Body Systems

A patient with sudden confusion, cool clammy skin, tremor, and tachycardia is found. What condition should be suspected and what must be checked first?

Answer

This presentation suggests hypoglycemia. Blood glucose measurement is mandatory whenever assessing a patient with altered level of consciousness, since hypoglycemia can mimic stroke, intoxication, or a psychiatric emergency.

#308 · Other Body Systems

How is severe hypoglycemia treated when a patient cannot safely swallow?

Answer

Paramedics may administer glucose intravenously or, where authorized, glucagon intramuscularly, which mobilizes the patient's own stored glycogen to raise blood glucose.

#309 · Other Body Systems

Describe the presentation of a thyroid storm.

Answer

A thyroid storm is a rare but severe complication of untreated or poorly controlled hyperthyroidism, presenting with high fever, severe tachycardia, agitation, and altered mental status, managed supportively with cooling, oxygen, and rapid transport.

#310 · Other Body Systems

How does myxedema coma present, and what causes it?

Answer

Myxedema coma is a life-threatening complication of severe untreated hypothyroidism, presenting with profound lethargy, hypothermia, bradycardia, and hypoventilation.

#311 · Other Body Systems

What is the difference between the compact and spongy layers of bone?

Answer

The compact layer is the hard outer layer of bone, while the spongy inner layer contains marrow, where blood cells are produced.

#312 · Other Body Systems

Distinguish between a closed fracture and an open fracture.

Answer

A closed fracture leaves the overlying skin intact, while an open fracture involves a break in the skin at or near the fracture site, creating a direct pathway for infection and significant blood loss.

#313 · Other Body Systems

Explain the difference between a sprain and a strain.

Answer

A sprain is an injury to a ligament from overstretching or tearing, commonly at the ankle or wrist, while a strain is an injury to a muscle or tendon from overstretching or overexertion.

#314 · Other Body Systems

What is the general splinting rule for a suspected fracture versus a suspected joint injury?

Answer

For a suspected fracture, immobilize the joint above and the joint below the injury. For a suspected joint injury, immobilize the bone above and the bone below, so the entire injured segment is stabilized.

#315 · Other Body Systems

When might local protocol direct a paramedic to gently realign an angulated limb before splinting?

Answer

If the limb is significantly angulated and distal circulation is absent, gentle realignment may be attempted to restore perfusion before splinting, since a pulseless limb left uncorrected risks permanent tissue damage.

#316 · Other Body Systems

What injury is a traction splint specifically used for, and why?

Answer

A traction splint is used for a suspected mid-shaft femur fracture. It applies a controlled pulling force to realign the bone, reduce pain, and limit ongoing blood loss into the thigh from the femoral vessels.

#317 · Other Body Systems

What causes compartment syndrome, and why can a distal pulse still be present despite it?

Answer

Bleeding or swelling inside a fascia-bound muscle compartment raises pressure until it exceeds capillary perfusion pressure, cutting off blood flow to muscles and nerves within that compartment, even though a larger distal artery may still produce a palpable pulse.

#318 · Other Body Systems

What are the classic signs of compartment syndrome?

Answer

Pain disproportionate to the apparent injury that worsens with passive stretching of the affected muscles, along with paresthesia, pallor, and a tense, swollen feeling to the compartment.

#319 · Other Body Systems

Name the three layers of the skin from outermost to deepest.

Answer

The epidermis (outermost, waterproof barrier with pigment-producing cells), the dermis (blood vessels, nerve endings, hair follicles, and sweat glands), and the subcutaneous tissue or hypodermis (fat and connective tissue that cushions and insulates).

#320 · Other Body Systems

Differentiate an abrasion from an avulsion.

Answer

An abrasion is superficial scraping away of the epidermis, often from sliding against a rough surface. An avulsion is a more severe injury where a flap or section of tissue is partially or completely torn away from its underlying structure.

#321 • Other Body Systems

How should a completely detached body part, such as an amputated fingertip, be managed?

Answer

It should be wrapped, kept cool, and transported with the patient whenever possible, in case reattachment is feasible at hospital.

#322 • Other Body Systems

What is the first-line technique for controlling significant external bleeding, and why should the dressing not be repeatedly lifted to check it?

Answer

Direct pressure applied firmly over the wound, ideally with a dressing, maintained continuously. Lifting the dressing to check disrupts early clot formation.

#323 · Other Body Systems

How should a tourniquet be applied for life-threatening extremity hemorrhage?

Answer

Applied high and tight above the injury, tightened until bleeding stops and the distal pulse is eliminated, with the time of application clearly documented for the receiving hospital.

#324 · Other Body Systems

When are hemostatic dressings used instead of a tourniquet?

Answer

Hemostatic dressings are used for wounds in junctional areas where a tourniquet cannot be applied, such as the groin, axilla, or neck, and are packed directly into the wound with continued direct pressure on top.

#325 · Other Body Systems

What is the correct field management of an abdominal evisceration?

Answer

Eviscerated organs should never be pushed back into the abdomen. They are covered with a moist sterile dressing to prevent drying, then an occlusive layer to retain moisture and warmth, with the patient kept warm, knees flexed to reduce abdominal wall tension, and transported promptly.

#326 · Other Body Systems

How should a paramedic manage an impaled object in a penetrating wound?

Answer

The object should be stabilized in place and never removed in the field, since it may be tamponading a bleeding vessel.

#327 · Other Body Systems

Distinguish hematemesis from melena.

Answer

Hematemesis is vomiting of blood that may appear bright red or resemble coffee grounds if partially digested. Melena is black tarry stool caused by digested blood passing through the bowel, both signs of upper GI bleeding.

#328 · Other Body Systems

What does hematochezia suggest about the source of GI bleeding?

Answer

Hematochezia, bright red blood passed rectally, more often suggests lower GI bleeding, arising from the lower small intestine, colon, or rectum.

#329 · Other Body Systems

Describe the classic pain pattern of appendicitis.

Answer

Appendicitis classically begins as vague periumbilical pain that migrates and localizes to the right lower quadrant, often accompanied by fever, nausea, and rebound tenderness.

#330 · Other Body Systems

What is the typical presentation of cholecystitis, and what commonly triggers it?

Answer

Right upper quadrant pain that may radiate to the right shoulder, often triggered or worsened after a fatty meal, typically caused by gallstones obstructing the gallbladder's outlet.

#331 · Other Body Systems

How does pancreatitis typically present?

Answer

Severe epigastric pain that frequently radiates straight through to the back, along with nausea and vomiting, most commonly related to gallstones or alcohol use.

#332 · Other Body Systems

Why must a patient with blunt abdominal trauma and a concerning mechanism be treated as having a serious internal injury even if the abdomen looks unremarkable?

Answer

Blunt abdominal trauma can cause internal organ injury and significant internal bleeding without any obvious external sign, so tenderness, distension, or signs of shock must be taken seriously regardless of external appearance.

#333 · Other Body Systems

What is the basic function of a nephron within the kidney?

Answer

The nephron is the kidney's tiny functional filtering unit, reabsorbing water and needed substances from the blood while excreting waste as urine.

#334 · Other Body Systems

Contrast the onset and typical causes of acute kidney injury versus chronic kidney disease.

Answer

Acute kidney injury develops rapidly over hours to days, often from severe dehydration, hypotension, toxins, or urine flow obstruction, and is frequently reversible. Chronic kidney disease develops gradually over months to years, most often from long-standing diabetes or hypertension, and is largely irreversible.

#335 · Other Body Systems

What is an arteriovenous fistula, and why is it created for dialysis patients?

Answer

An arteriovenous fistula is a surgically created connection between an artery and vein, usually in the arm, that provides a durable, high-flow access point for repeated needle insertion during hemodialysis.

#336 · Other Body Systems

What two precautions must a paramedic follow when the arm has a dialysis fistula or graft?

Answer

Blood pressure should never be measured and intravenous access should never be attempted on the arm with a fistula or graft, since any pressure or needle puncture risks damaging this difficult-to-replace access point.

#337 · Other Body Systems

What ECG changes suggest dangerous hyperkalemia in a dialysis patient?

Answer

Peaked T waves or a widened QRS complex on the cardiac monitor are characteristic changes that should prompt paramedics to prioritize rapid transport and continuous cardiac monitoring.

#338 · Other Body Systems

How does peritoneal dialysis differ from hemodialysis?

Answer

Peritoneal dialysis uses the patient's own peritoneal membrane inside the abdomen as a filter, with dialysis fluid instilled and drained through a permanent abdominal catheter, whereas hemodialysis filters blood through an external machine using vascular access such as a fistula or graft.

#339 · Other Body Systems

Why is testicular torsion considered a true urologic emergency requiring prompt transport?

Answer

The testicle rotates on the spermatic cord, cutting off its own blood supply, and it can become permanently nonviable within hours if blood flow is not restored, so unnecessary delay or invasive field examination should be avoided.

#340 · Other Body Systems

What is priapism and what conditions can cause it?

Answer

Priapism is a prolonged, often painful erection unrelated to sexual stimulation, which can result from certain medications, sickle cell disease, or spinal cord injury, and is time sensitive due to risk of lasting tissue damage.

#341 · Other Body Systems

What is an ectopic pregnancy and why is it dangerous?

Answer

An ectopic pregnancy occurs when a fertilized egg implants outside the uterus, most commonly within a fallopian tube, which cannot safely accommodate a growing pregnancy. Rupture can cause rapid, life-threatening internal hemorrhage and shock.

#342 · Other Body Systems

What is ovarian torsion and how does it compare to testicular torsion?

Answer

Ovarian torsion occurs when an ovarian cyst causes the ovary to twist on its blood supply, producing sudden sharp pelvic pain. It mirrors testicular torsion in both its urgency and its mechanism.

#343 · Other Body Systems

How does pelvic inflammatory disease typically present, and what commonly causes it?

Answer

Pelvic inflammatory disease presents with lower abdominal pain, fever, and abnormal vaginal discharge, often arising from an untreated sexually transmitted infection, and can progress to abscess formation and infertility if untreated.

#344 · Other Body Systems

What should paramedics gather when assessing any patient of reproductive age with abdominal pain?

Answer

A focused history including onset, character, and location of pain, associated bleeding or discharge, and pregnancy status where relevant, while maintaining patient privacy and dignity throughout.

#345 · Other Body Systems

What unifying field management principle applies across the reproductive emergencies discussed in this chapter, such as torsion and ectopic pregnancy?

Answer

Management is largely supportive: monitoring for signs of shock, providing oxygen and fluid support per protocol, positioning for comfort, and transporting promptly, since definitive diagnosis and treatment can only occur in hospital.

#346 · Trauma Toxicology and Environmental Emergencies

What should a paramedic do before approaching a suspected poisoning scene with an unlabelled container or a strong chemical odour?

Answer

Stage at a safe distance and call for additional resources rather than walking directly in, since the substance could also harm the responding crew.

#347 · Trauma Toxicology and Environmental Emergencies

Name the four main routes by which a poison can enter the body.

Answer

Ingestion, inhalation, injection, and absorption through intact skin or mucous membranes.

#348 · Trauma Toxicology and Environmental Emergencies

Why does inhalation of a toxic gas or vapour typically cause faster onset of symptoms than ingestion of a pill?

Answer

The lungs offer a large alveolar absorption surface, so an inhaled substance is absorbed rapidly, while an ingested substance must first pass through the gastrointestinal tract, delaying onset.

#349 · Trauma Toxicology and Environmental Emergencies

Which route of exposure carries a real risk of secondary contamination to responders who touch a patient's skin without protection?

Answer

Absorption, as seen with organophosphate pesticides that remain on the skin and can transfer to bare hands or clothing.

#350 · Trauma Toxicology and Environmental Emergencies

What key history points should be gathered on any suspected poisoning or overdose call?

Answer

What was taken, how much, when it was taken, whether alcohol or other substances were involved, and whether the exposure was intentional.

#351 · Trauma Toxicology and Environmental Emergencies

Define a toxidrome.

Answer

A recognizable cluster of signs and symptoms produced by a class of substances, allowing a paramedic to anticipate complications before the exact substance is identified.

#352 · Trauma Toxicology and Environmental Emergencies

Describe the classic triad of the opioid toxidrome.

Answer

Pinpoint pupils, decreased level of consciousness, and respiratory depression that can progress to apnea, with skin often cool and clammy.

#353 · Trauma Toxicology and Environmental Emergencies

Which toxidrome will an AEMCA encounter most often, and what substances typically cause it?

Answer

The opioid toxidrome, seen with heroin, fentanyl, oxycodone, and morphine.

#354 · Trauma Toxicology and Environmental Emergencies

What clinical picture defines the sympathomimetic toxidrome, and which drugs cause it?

Answer

Tachycardia, hypertension, dilated pupils, agitation, diaphoresis, and hyperthermia, associated with cocaine and methamphetamine, with severe cases progressing to seizures or dysrhythmias.

#355 · Trauma Toxicology and Environmental Emergencies

Recite the memory aid for the anticholinergic toxidrome and its meaning.

Answer

Hot as a hare, blind as a bat, dry as a bone, red as a beet, and mad as a hatter: elevated temperature, dilated poorly reactive pupils, dry flushed skin, and delirium.

#356 · Trauma Toxicology and Environmental Emergencies

Which substances typically produce an anticholinergic toxidrome?

Answer

Certain antihistamines and tricyclic antidepressants.

#357 · Trauma Toxicology and Environmental Emergencies

What does the SLUDGE mnemonic stand for in the cholinergic toxidrome?

Answer

Salivation, lacrimation, urination, defecation, gastrointestinal upset, and emesis.

#358 · Trauma Toxicology and Environmental Emergencies

What causes the cholinergic toxidrome, and why does it make suction a top priority?

Answer

It is caused by excess acetylcholine activity, typically from organophosphate pesticides or nerve agents, and it can rapidly compromise the airway through excess secretions and bronchospasm.

#359 · Trauma Toxicology and Environmental Emergencies

Besides bronchorrhea and bronchospasm, what heart rate finding is typical of the cholinergic toxidrome?

Answer

Bradycardia.

#360 · Trauma Toxicology and Environmental Emergencies

Why is acetaminophen overdose considered dangerous despite the patient often looking well early on?

Answer

The threat is delayed hepatotoxicity progressing to liver failure over one to several days, so a reported acetaminophen overdose should always be treated as a genuine emergency regardless of appearance.

#361 · Trauma Toxicology and Environmental Emergencies

What clinical pattern should raise suspicion for salicylate (ASA) toxicity?

Answer

Tinnitus, nausea, vomiting, and rapid, deep hyperventilation as the body compensates for a developing metabolic acidosis, with confusion, seizures, or cardiovascular collapse possible as toxicity progresses.

#362 · Trauma Toxicology and Environmental Emergencies

Why can alcohol intoxication be dangerous to assume as the only explanation for a patient's presentation?

Answer

A patient who appears simply intoxicated may actually be hypoglycemic, have a head injury from a fall, or have co-ingested another substance, so a thorough assessment should not attribute every finding to intoxication alone.

#363 · Trauma Toxicology and Environmental Emergencies

How does alcohol contribute to hypothermia?

Answer

It causes peripheral vasodilation, which increases heat loss from the body, and it also impairs the gag reflex.

#364 · Trauma Toxicology and Environmental Emergencies

What is the prehospital management priority for opioid overdose, and how does naloxone fit into it?

Answer

Supporting oxygenation and ventilation, including bag valve mask ventilation if breathing is inadequate, combined with naloxone according to local medical directives; naloxone is an opioid antagonist that displaces opioids from their receptors and reverses respiratory depression within a few minutes.

#365 · Trauma Toxicology and Environmental Emergencies

Why can a patient re-sedate after receiving naloxone, and what other reaction can naloxone precipitate?

Answer

Naloxone's duration of action may be shorter than the opioid involved, so the patient can re-sedate once it wears off; naloxone can also precipitate acute withdrawal and combativeness.

#366 · Trauma Toxicology and Environmental Emergencies

Why is carbon monoxide poisoning easy to miss on scene?

Answer

The gas is colourless and odourless, and symptoms such as headache, nausea, dizziness, and confusion are nonspecific, though multiple simultaneously affected people in the same household or vehicle is an important clue.

#367 · Trauma Toxicology and Environmental Emergencies

Why can standard pulse oximetry be misleading in carbon monoxide poisoning?

Answer

It cannot reliably distinguish carbon monoxide-bound hemoglobin from oxygen-bound hemoglobin, so it can give a falsely reassuring reading.

#368 · Trauma Toxicology and Environmental Emergencies

What is the mainstay of prehospital management for carbon monoxide poisoning?

Answer

High-concentration oxygen therapy, since it accelerates removal of carbon monoxide from hemoglobin.

#369 · Trauma Toxicology and Environmental Emergencies

Is inducing vomiting part of prehospital management for most poisoning calls?

Answer

No, inducing vomiting is not part of prehospital management for most ingestions.

#370 · Trauma Toxicology and Environmental Emergencies

How does heat exhaustion typically present, including mental status and skin findings?

Answer

Heavy sweating, weakness, nausea, headache, and dizziness, with skin often cool, pale, and clammy, core temperature normal or only mildly elevated, and mental status largely normal.

#371 · Trauma Toxicology and Environmental Emergencies

What is the prehospital management for heat exhaustion?

Answer

Remove the patient from the hot environment, cool with fanning and cool water, loosen restrictive clothing, and provide rest and oral fluids if the patient is alert and tolerating them.

#372 · Trauma Toxicology and Environmental Emergencies

What is the hallmark finding that distinguishes heat stroke from heat exhaustion?

Answer

Altered mental status, ranging from confusion to seizures to unresponsiveness, combined with significantly elevated core temperature.

#373 · Trauma Toxicology and Environmental Emergencies

Why can skin condition alone not rule out heat stroke?

Answer

Skin may be hot and either dry or still moist, since sweating does not reliably stop in every case of heat stroke.

#374 · Trauma Toxicology and Environmental Emergencies

What is the priority management for heat stroke?

Answer

Aggressive cooling: remove clothing, apply cool water or ice packs to the neck, axillae, and groin, and transport rapidly, since prolonged hyperthermia damages the brain and kidneys.

#375 · Trauma Toxicology and Environmental Emergencies

How does a patient with mild hypothermia typically present?

Answer

Shivering, alert, and able to compensate, though coordination and judgment may begin to decline.

#376 · Trauma Toxicology and Environmental Emergencies

What changes occur as hypothermia progresses from mild to moderate?

Answer

Shivering diminishes or stops, consciousness declines further, and the patient becomes lethargic and uncoordinated with slowed speech.

#377 · Trauma Toxicology and Environmental Emergencies

What can severe hypothermia mimic, and why?

Answer

Clinical death, because shivering has ceased, the patient may be unresponsive, and pulse and respirations can become extremely slow and difficult to detect.

#378 · Trauma Toxicology and Environmental Emergencies

Why must a severely hypothermic patient be handled gently and moved smoothly?

Answer

The heart becomes highly irritable in severe hypothermia, so rough handling, jostling, or aggressive movement can precipitate a lethal dysrhythmia such as ventricular fibrillation.

#379 · Trauma Toxicology and Environmental Emergencies

What is the correct field approach to protecting a hypothermic patient from further heat loss?

Answer

Remove wet clothing and replace it with dry insulation, keep the patient horizontal, and avoid aggressive rewarming or vigorous rubbing in the field.

#380 · Trauma Toxicology and Environmental Emergencies

What are the early signs of frostbite?

Answer

Numbness, pallor, and a firm or waxy feel to the skin, most commonly in the fingers, toes, ears, and nose.

#381 · Trauma Toxicology and Environmental Emergencies

Why is field rewarming of frostbite generally avoided if there is a risk of refreezing?

Answer

A freeze-thaw-refreeze cycle causes more tissue damage than a single sustained freeze.

#382 · Trauma Toxicology and Environmental Emergencies

What is the central problem driving every downstream complication in drowning, including cardiac arrest?

Answer

Hypoxia, which develops quickly once water or laryngospasm interrupts normal gas exchange.

#383 · Trauma Toxicology and Environmental Emergencies

What is the priority once a drowning patient is safely removed from the water by trained rescuers?

Answer

Opening the airway and providing rescue breaths or bag valve mask ventilation as soon as possible, followed by chest compressions if the patient is in cardiac arrest.

#384 · Trauma Toxicology and Environmental Emergencies

Should airway and ventilation be delayed for spinal precautions in a drowning patient with a suspected shallow-water dive injury?

Answer

No, cervical spine injury should be considered, but airway and ventilation should never be delayed for spinal precautions.

#385 · Trauma Toxicology and Environmental Emergencies

Why should all drowning patients be transported, even those who recover quickly?

Answer

Pulmonary complications can develop hours later even after apparent recovery.

#386 · Trauma Toxicology and Environmental Emergencies

What is barotrauma, and what serious complication can pulmonary barotrauma cause?

Answer

Barotrauma is tissue injury from pressure differences between an air-filled body space and the surrounding environment; pulmonary barotrauma from ascending too quickly while holding the breath can force air into the bloodstream, producing an arterial gas embolism with rapid, severe neurological symptoms.

#387 · Trauma Toxicology and Environmental Emergencies

What causes decompression sickness, and what symptoms can it produce?

Answer

Nitrogen bubbles forming in blood and tissue when a diver ascends too quickly after breathing compressed air at depth, producing joint pain, fatigue, skin changes, and in severe cases neurological deficits, sometimes delayed for hours after surfacing.

#388 · Trauma Toxicology and Environmental Emergencies

What is the prehospital management for suspected decompression sickness or arterial gas embolism?

Answer

High-concentration oxygen and prompt transport to a facility capable of definitive treatment.

#389 · Trauma Toxicology and Environmental Emergencies

What is acute mountain sickness, and how is it managed?

Answer

The mildest and most common form of altitude illness, presenting with headache, fatigue, nausea, dizziness, and difficulty sleeping within a day of arriving at altitude, managed by halting further ascent, allowing acclimatization, and descending if symptoms worsen.

#390 · Trauma Toxicology and Environmental Emergencies

What distinguishes high-altitude pulmonary edema from acute mountain sickness?

Answer

It involves fluid in the lungs, with worsening shortness of breath, cough, and reduced exercise tolerance, sometimes pink frothy sputum, representing a more serious progression than acute mountain sickness.

#391 · Trauma Toxicology and Environmental Emergencies

What is the definitive treatment for high-altitude pulmonary edema and high-altitude cerebral edema?

Answer

Descent as soon as it can be safely arranged, alongside supplemental oxygen.

#392 · Trauma Toxicology and Environmental Emergencies

What is the prehospital priority for a thermal burn, and what should be avoided during cooling?

Answer

Stopping the burning process and cooling with room-temperature water, avoiding ice or prolonged cooling that can worsen injury or contribute to hypothermia.

#393 · Trauma Toxicology and Environmental Emergencies

What is the correct approach to a chemical burn, and why should dry chemical powders be brushed away first?

Answer

Remove contaminated clothing and irrigate with copious water for an extended period; dry chemical powders should generally be brushed away before irrigation, since water can trigger a reaction with certain dry chemicals.

#394 · Trauma Toxicology and Environmental Emergencies

Why can electrical burns be more dangerous than they appear on the surface?

Answer

They can look deceptively minor on the skin while causing extensive deep tissue damage along the current's path, and they carry a real risk of cardiac dysrhythmia, so cardiac monitoring is important.

#395 · Trauma Toxicology and Environmental Emergencies

Describe the appearance of a superficial burn.

Answer

Involves only the outer skin layer, appearing red, painful, and dry without blistering, such as a sunburn.

#396 · Trauma Toxicology and Environmental Emergencies

Describe the appearance of a partial-thickness burn and why it remains very painful.

Answer

It extends into the deeper skin layer, appearing red, moist, blistered, and quite painful, since nerve endings remain intact.

#397 · Trauma Toxicology and Environmental Emergencies

Describe the appearance of a full-thickness burn and explain why the deepest areas may be painless.

Answer

It extends through the entire skin and can involve fat, muscle, or bone, appearing dry, leathery, or waxy, and may be surprisingly painless in the deepest areas because nerve endings are destroyed.

#398 · Trauma Toxicology and Environmental Emergencies

Under the rule of nines, what body surface area percentage is assigned to each entire arm and to the head and neck in an adult?

Answer

Each entire arm is approximately nine percent, and the head and neck are approximately nine percent.

#399 · Trauma Toxicology and Environmental Emergencies

Under the rule of nines, what body surface area percentage is assigned to the anterior trunk, the posterior trunk, and each leg in an adult?

Answer

The anterior trunk is approximately eighteen percent, the posterior trunk is approximately eighteen percent, and each entire leg is approximately eighteen percent.

#400 · Trauma Toxicology and Environmental Emergencies

Why does a child's rule of nines chart differ from an adult's?

Answer

A child's head is relatively larger than an adult's, so an adjusted pediatric chart is used to estimate body surface area burned.

#401 · Trauma Toxicology and Environmental Emergencies

What warning signs suggest an airway burn in a patient burned by fire or smoke?

Answer

Singed facial hair, soot around the mouth or nose, hoarseness, and difficulty swallowing, all of which warrant close monitoring and prompt transport since airway swelling can progress rapidly and unpredictably.

#402 · Trauma Toxicology and Environmental Emergencies

What is the prehospital focus for a significant burn, given that fluid resuscitation is primarily an in-hospital consideration?

Answer

Covering the burn with a clean, dry dressing, preventing further contamination and heat loss, providing analgesia within scope, and transporting promptly.

#403 · Trauma Toxicology and Environmental Emergencies

Why can internal injury from a gunshot wound be far more extensive than the visible external wound suggests?

Answer

A temporary cavity can form as the projectile passes through tissue, causing injury beyond the visible track, influenced by the projectile's velocity, path, and whether it tumbles, fragments, or strikes bone or a major vessel.

#404 · Trauma Toxicology and Environmental Emergencies

Why should wound appearance never be used to determine firing direction in a suspected gunshot wound?

Answer

An entrance wound is often smaller and more regular while an exit wound can be larger and more irregular, but this pattern is not absolute, so it should never be relied on to determine firing direction.

#405 · Trauma Toxicology and Environmental Emergencies

How should an impaled object generally be managed in the field, and why?

Answer

It should generally be stabilized in place rather than removed, since the object may be tamponading a bleeding vessel.

#406 · Shock Fluids and Medical Emergencies

Why should a paramedic never assume a behavioural presentation is purely psychiatric?

Answer

A behavioural presentation is a symptom, not a diagnosis, and organic causes such as hypoglycemia, hypoxia, head injury, stroke, sepsis, drug intoxication, alcohol withdrawal, or thyroid storm can all mimic agitation or confusion. A blood glucose check, oxygen saturation reading, and basic neurological assessment belong on every behavioural call before a psychiatric explanation is settled on.

#407 · Shock Fluids and Medical Emergencies

What should a paramedic do before approaching a scene involving altered behaviour with a possible weapon?

Answer

Stage at a safe distance until police confirm the scene is secure. Once on scene, identify at least two exit routes, secure obvious hazards within the patient's reach, and avoid turning your back on an agitated individual.

#408 · Shock Fluids and Medical Emergencies

List three effective de-escalation techniques for an agitated patient.

Answer

Approach slowly while introducing yourself and stating your purpose, use the patient's name and short simple sentences, and give the patient space rather than crowding them. A calm, even voice at a slower pace than the patient's own can lower the emotional temperature of the encounter.

#409 · Shock Fluids and Medical Emergencies

How should a paramedic respond to a patient who is actively experiencing a delusion or hallucination?

Answer

Never argue with the delusion or try to talk the patient out of the hallucination, since this often increases agitation. Instead, acknowledge the patient's distress without confirming or denying the delusion, and redirect toward a concrete, answerable question.

#410 · Shock Fluids and Medical Emergencies

Does apprehension under the Mental Health Act change how a paramedic clinically manages a patient?

Answer

No. The clinical role is unchanged: continue monitoring vital signs, maintain a calm and respectful approach, and document objective behavioural observations rather than subjective labels. An apprehended patient still deserves full clinical assessment.

#411 · Shock Fluids and Medical Emergencies

Name several warning signs of suicide risk a paramedic should recognize.

Answer

Warning signs include direct statements of intent, a specific plan, access to means, recent major loss, previous attempts, giving away possessions, and sudden calm after severe depression, which can indicate a decision to act.

#412 · Shock Fluids and Medical Emergencies

What is the paramedic's actual role when a patient expresses suicidal intent?

Answer

Not to conduct a formal risk assessment, but to recognize red flags, maintain safety during transport, and communicate findings clearly to receiving hospital staff.

#413 · Shock Fluids and Medical Emergencies

What triggers the fight-or-flight response and what chemicals mediate it?

Answer

The sympathetic nervous system triggers the response through release of catecholamines such as epinephrine and norepinephrine, producing increased heart rate and respiratory rate, pupil dilation, and redirection of blood flow away from the digestive system toward skeletal muscle.

#414 · Shock Fluids and Medical Emergencies

Why can a patient's own stress response complicate paramedic assessment?

Answer

A patient's anxiety can mimic or mask symptoms of a genuine medical problem, so recognizing a stress response for what it is, rather than treating it as the primary complaint, is an important assessment skill.

#415 · Shock Fluids and Medical Emergencies

What does each letter of the OPQRST pain assessment mnemonic stand for?

Answer

Onset, Provocation and Palliation, Quality, Region and Radiation, Severity, and Time. It is the standard framework for systematically assessing pain.

#416 · Shock Fluids and Medical Emergencies

How should pain be assessed in a patient who cannot communicate verbally?

Answer

Use a behavioural or observational pain scale based on facial expression, body movement, and vocalization instead of a numeric self-report.

#417 · Shock Fluids and Medical Emergencies

Give three non-pharmacological pain management measures within PCP scope.

Answer

Splinting an injured limb, positioning the patient comfortably, and applying cold packs where appropriate, along with providing reassurance, all reduce pain perception even without medication.

#418 · Shock Fluids and Medical Emergencies

Why is reassessment important after administering an analgesic under a local medical directive?

Answer

Established protocols require a contraindication checklist and a reassessment requirement after administration, since the exam expects understanding of the general approach to evaluating whether pain relief was effective rather than a specific drug name or dose.

#419 · Shock Fluids and Medical Emergencies

What signs suggest a paramedic colleague may be experiencing cumulative stress injury?

Answer

Irritability, sleep disturbance, and emotional withdrawal are early signs. Peer support and critical incident stress management resources exist to address this real occupational risk.

#420 · Shock Fluids and Medical Emergencies

What hormones help regulate fluid and electrolyte balance in the body?

Answer

Antidiuretic hormone and aldosterone, working alongside the kidneys and cardiovascular system, help maintain the tightly regulated balance of water and electrolytes between intracellular and extracellular compartments.

#421 · Shock Fluids and Medical Emergencies

How does severe dehydration present differently from mild dehydration?

Answer

Mild dehydration shows only thirst and slightly dry mucous membranes, while severe dehydration produces tachycardia, hypotension, delayed capillary refill, altered level of consciousness, and poor skin turgor, representing a true emergency requiring rapid transport.

#422 · Shock Fluids and Medical Emergencies

Which patient populations are at particularly high risk from dehydration, and why?

Answer

Elderly patients and infants are at highest risk because of reduced physiological reserve, meaning they compensate less effectively for fluid loss.

#423 · Shock Fluids and Medical Emergencies

What ECG changes suggest hyperkalemia in the field?

Answer

Peaked T waves, a widened QRS complex, and in severe cases a sine-wave pattern progressing to lethal dysrhythmia. A dialysis patient who has missed treatment presenting with weakness or cardiac irregularity should raise strong suspicion even without a lab value.

#424 · Shock Fluids and Medical Emergencies

What symptoms suggest hyponatremia, and what can cause it?

Answer

Confusion, headache, nausea, muscle cramps, and in severe or rapidly developing cases seizures and altered level of consciousness. It can result from excessive water intake, certain medications, or conditions causing inappropriate water retention.

#425 · Shock Fluids and Medical Emergencies

What is the paramedic's role regarding suspected electrolyte disturbances in the field?

Answer

Not to diagnose the specific abnormality, since lab work is unavailable in the field, but to recognize that unexplained cardiac irregularity or altered mental status in a patient with relevant history may reflect an electrolyte problem that changes urgency and destination decisions.

#426 · Shock Fluids and Medical Emergencies

Why is shock not defined by a single blood pressure number?

Answer

Shock is a physiological state of inadequate tissue perfusion that can exist even when blood pressure remains within a technically normal range, which is why understanding the stages of shock matters more than fixating on one vital sign.

#427 · Shock Fluids and Medical Emergencies

What subtle signs indicate a patient is in compensated shock despite a normal blood pressure?

Answer

Anxiety or restlessness, pale and cool skin, delayed capillary refill, and a rising heart rate. In this stage the body compensates through vasoconstriction, tachycardia, and catecholamine release, and early recognition here makes the greatest difference.

#428 · Shock Fluids and Medical Emergencies

What happens as a patient transitions from compensated to decompensated shock?

Answer

Blood pressure drops, level of consciousness deteriorates, skin becomes mottled or cyanotic, capillary refill is markedly delayed, and the patient may become unresponsive. This is a true emergency because the window for reversing the process narrows quickly once compensation fails.

#429 · Shock Fluids and Medical Emergencies

What are the management priorities for hypovolemic shock from hemorrhage?

Answer

Controlling external bleeding, positioning the patient supine, keeping the patient warm to prevent hypothermia, acidosis, and coagulopathy, providing supplemental oxygen, and rapid transport.

#430 · Shock Fluids and Medical Emergencies

What distinguishes cardiogenic shock from hypovolemic shock?

Answer

In cardiogenic shock the problem is not a lack of volume but a failure of the heart to pump the volume present, often following a large myocardial infarction, severe dysrhythmia, or acute heart failure. Patients show both poor perfusion and fluid backup, including pulmonary crackles, jugular venous distension, and peripheral edema.

#431 · Shock Fluids and Medical Emergencies

Why can aggressive fluid administration worsen cardiogenic shock?

Answer

Because the heart is already failing as a pump, adding more volume can overload it further, which is why distinguishing cardiogenic shock from hypovolemic shock matters clinically before considering fluid administration.

#432 · Shock Fluids and Medical Emergencies

What is the common feature shared by all forms of distributive shock?

Answer

Abnormal vasodilation causing a relative hypovolemia, where total blood volume may be normal but the vascular space has expanded beyond what it can fill.

#433 · Shock Fluids and Medical Emergencies

Why might an elderly patient in septic shock not present with fever?

Answer

Septic shock produces widespread vasodilation and capillary leak, and typically fever, but some septic patients, particularly the elderly, may present with a normal or low temperature instead.

#434 · Shock Fluids and Medical Emergencies

What is the definitive, time-critical treatment for anaphylactic shock, and how is it given?

Answer

Epinephrine administered by intramuscular injection into the anterolateral thigh. Delay in administration is associated with worse outcomes, and antihistamines and other adjuncts should never delay it when anaphylaxis is suspected.

#435 · Shock Fluids and Medical Emergencies

What is a biphasic reaction and why does it matter after treating anaphylaxis?

Answer

It is a second wave of anaphylactic symptoms that can occur hours after the reaction initially appears resolved, so patients who receive epinephrine require ongoing monitoring.

#436 · Shock Fluids and Medical Emergencies

Why does neurogenic shock often present with a normal or slow heart rate instead of tachycardia?

Answer

A spinal cord injury disrupts sympathetic output below the level of injury, interrupting the sympathetic drive that would normally raise heart rate, producing hypotension without compensatory tachycardia, a frequently tested distinguishing feature.

#437 · Shock Fluids and Medical Emergencies

Why are neurogenic shock patients at particular risk of hypothermia?

Answer

They lose the ability to shiver and vasoconstrict below the level of injury, so management includes spinal motion restriction as indicated, maintaining the airway, and actively preventing hypothermia.

#438 · Shock Fluids and Medical Emergencies

What defines obstructive shock, and name its three common causes.

Answer

A physical obstruction that prevents adequate blood flow into or out of the heart despite normal blood volume and cardiac muscle function. Common causes are tension pneumothorax, cardiac tamponade, and massive pulmonary embolism.

#439 · Shock Fluids and Medical Emergencies

What is the classic triad of cardiac tamponade, and is it always fully present?

Answer

Hypotension, jugular venous distension, and muffled heart sounds, though all three are not always present together in a given patient.

#440 · Shock Fluids and Medical Emergencies

Why can fluid administration alone not fix obstructive shock?

Answer

The definitive treatment is relieving the physical obstruction itself, something fluid cannot accomplish; tension pneumothorax may be addressed within a paramedic's scope while cardiac tamponade and massive pulmonary embolism require urgent hospital intervention.

#441 • Shock Fluids and Medical Emergencies

Why is fever in a chemotherapy patient always treated as a true emergency?

Answer

Chemotherapy often causes neutropenia, a dangerously low white blood cell count, and a neutropenic patient with fever, even low-grade, can progress from a minor infection to sepsis very quickly, so prompt transport is warranted.

#442 • Shock Fluids and Medical Emergencies

How should a paramedic handle a patient's implanted vascular access device, such as a chemotherapy port?

Answer

Avoid unnecessary manipulation of the device and clearly communicate its presence to receiving hospital staff, since it may be an appropriate treatment route and requires infection precautions specific to indwelling devices.

#443 · Shock Fluids and Medical Emergencies

What should a paramedic do when no valid Do Not Resuscitate documentation is present for a patient with advanced cancer?

Answer

Standard resuscitation guidelines apply even in a patient with known advanced illness, and the paramedic should never assume a patient's wishes based on diagnosis alone.

#444 · Shock Fluids and Medical Emergencies

Compare droplet and airborne transmission of communicable disease.

Answer

Droplet transmission occurs when respiratory droplets travel a short distance, typically less than two metres, and contact mucous membranes. Airborne transmission involves much smaller particles that remain suspended in air longer and travel farther, requiring a higher level of respiratory protection such as a fit-tested N95 respirator.

#445 · Shock Fluids and Medical Emergencies

What steps should a paramedic take immediately after a needlestick injury or significant unprotected exposure?

Answer

Follow the service's post-exposure protocol: immediate first aid such as washing the affected area, prompt reporting to a supervisor, medical evaluation, and documentation, since timely reporting affects both health monitoring and any indicated post-exposure prophylaxis.

#446 · Special Populations

What marks the beginning and end of the first stage of labour?

Answer

The first stage begins with the onset of regular contractions and ends with full cervical dilation to 10 centimetres. It is typically the longest stage and includes a latent phase with irregular contractions and an active phase with stronger, closer, more painful contractions.

#447 · Special Populations

What happens during the second and third stages of labour?

Answer

The second stage begins at full dilation and ends with delivery of the baby (the pushing stage). The third stage begins right after the baby is delivered and ends with delivery of the placenta, usually within 30 minutes.

#448 · Special Populations

In an obstetrical history, what do gravida and para mean?

Answer

Gravida is the total number of pregnancies a patient has had, and para is the total number of deliveries. Both help you gauge how quickly labour may progress.

#449 · Special Populations

Which findings tell you delivery is imminent and you should prepare to deliver on scene?

Answer

Crowning, an uncontrollable urge to push, or a visible presenting part all mean delivery is imminent. In these cases you should prepare to deliver on scene rather than attempt transport.

#450 · Special Populations

How should you position a labouring patient once delivery is imminent?

Answer

Position the patient supine with the knees bent and support under the hips if possible, maintain privacy, and prepare a clean field with the obstetrical kit.

#451 · Special Populations

What is a nuchal cord and how is it managed during delivery of the head?

Answer

A nuchal cord is an umbilical cord wrapped around the newborn's neck. As the head delivers, check for it and, if present, attempt to gently slip it over the head.

#452 · Special Populations

When should you suction a newborn's mouth and nose during delivery?

Answer

Suction the mouth before the nose only if the newborn shows obvious airway obstruction from secretions. Routine suctioning of a vigorous newborn is no longer recommended.

#453 · Special Populations

How is the umbilical cord clamped and cut after delivery?

Answer

Clamp the cord in two places roughly 10 and 15 centimetres from the newborn, keeping the newborn at the level of the placenta until clamped, then cut between the clamps with sterile scissors.

#454 · Special Populations

Why should you never pull on the umbilical cord to speed delivery of the placenta?

Answer

Pulling on the cord to hasten the third stage can cause uterine inversion or hemorrhage. The placenta should be allowed to deliver on its own.

#455 · Special Populations

Why must the delivered placenta be transported to hospital with the patient?

Answer

The receiving team will inspect the placenta for completeness, because retained placental fragments are a leading cause of postpartum hemorrhage.

#456 · Special Populations

What is uterine massage and why is it performed after delivery?

Answer

Uterine massage is rubbing the fundus firmly in a circular motion through the lower abdomen. It helps the uterus contract and reduces postpartum bleeding.

#457 • Special Populations

How is pre-eclampsia defined, and what symptoms often accompany it?

Answer

Pre-eclampsia is new-onset hypertension after 20 weeks gestation, often with protein in the urine, headache, visual disturbances, upper abdominal pain, and swelling of the hands and face.

#458 • Special Populations

What is eclampsia, and how is it managed prehospitally?

Answer

Eclampsia is the progression of pre-eclampsia to generalized seizures. Management includes airway protection, positioning the patient on her left side to relieve pressure on the inferior vena cava, minimizing stimulation, and rapid transport.

#459 · Special Populations

How does placenta previa typically present?

Answer

Placenta previa occurs when the placenta implants low in the uterus and covers or approaches the cervix, producing painless bright red vaginal bleeding in the third trimester.

#460 · Special Populations

How does abruptio placentae typically present, and why can it be deceptively dangerous?

Answer

Abruptio placentae is premature separation of the placenta from the uterine wall, causing painful bleeding that may be dark, a rigid or tender abdomen, and fetal distress. Because hemorrhage can be concealed behind the placenta, visible blood loss may understate true severity.

#461 • Special Populations

Why should you never perform a vaginal examination on a patient with third-trimester bleeding?

Answer

A vaginal examination in the prehospital setting can worsen bleeding from either placenta previa or abruptio placentae, so it should never be performed.

#462 • Special Populations

What is a prolapsed umbilical cord and why is it dangerous?

Answer

A prolapsed cord occurs when the umbilical cord slips through the cervix ahead of the presenting part, risking cord compression and fetal hypoxia with every contraction.

#463 · Special Populations

How do you manage a prolapsed cord in the field?

Answer

Position the mother knee-chest or in steep Trendelenburg, insert a gloved hand into the vagina to gently lift the presenting part off the cord without pushing the cord back in, keep the cord warm and moist, and transport immediately with the hand maintained in position.

#464 · Special Populations

What is breech presentation and how should an in-progress breech delivery be handled?

Answer

Breech presentation means the buttocks or feet present first. If delivery cannot be stopped, allow the body to deliver on its own without pulling, since the head, being largest, takes longer to deliver last.

#465 · Special Populations

What emergency airway manoeuvre is described for a breech delivery where the head fails to deliver and the newborn shows airway compromise?

Answer

Two fingers can be inserted into the vagina to form a V on either side of the newborn's nose and mouth to create an airway. This is an advanced manoeuvre reserved for genuine emergencies, ideally with online medical direction guidance.

#466 · Special Populations

How is postpartum hemorrhage defined, and what are its main causes?

Answer

Postpartum hemorrhage is blood loss exceeding roughly 500 millilitres after vaginal delivery. Causes include uterine atony, retained placental tissue, and lacerations of the birth canal.

#467 • Special Populations

How is postpartum hemorrhage managed in the field?

Answer

Management includes uterine fundal massage, encouraging breastfeeding if conditions allow (since this stimulates natural oxytocin release), treating for shock with oxygen and positioning, and rapid transport.

#468 • Special Populations

What is an ectopic pregnancy and how does it typically present when it ruptures?

Answer

An ectopic pregnancy occurs when a fertilized egg implants outside the uterine cavity, most often in a fallopian tube. Rupture can cause sudden severe unilateral lower abdominal pain, vaginal bleeding, and signs of internal hemorrhage and shock.

#469 • Special Populations

Which patients should raise suspicion for ectopic pregnancy?

Answer

Any patient of reproductive age with unexplained abdominal pain, syncope, or signs of shock should raise suspicion, even before she knows she is pregnant. Treatment is supportive: oxygen, shock positioning, IV access if within scope, and rapid transport to a facility capable of surgical intervention.

#470 • Special Populations

What are the initial steps immediately after a newborn is born?

Answer

Dry the newborn thoroughly, keep the newborn warm, position the airway neutrally, and stimulate by rubbing the back or flicking the soles of the feet.

#471 • Special Populations

Why do newborns lose heat so quickly after birth?

Answer

Newborns have a large surface-area-to-volume ratio, which causes rapid heat loss, making warming a priority immediately after delivery.

#472 • Special Populations

When does a newborn require positive pressure ventilation after the initial steps?

Answer

A newborn who is not breathing adequately, remains limp, or has a heart rate below 100 beats per minute after the initial steps requires positive pressure ventilation with an appropriately sized bag-valve-mask.

#473 • Special Populations

When are chest compressions added during neonatal resuscitation?

Answer

Chest compressions are added if the newborn's heart rate remains below 60 beats per minute despite effective ventilation for 30 seconds.

#474 • Special Populations

What does the APGAR concept evaluate, and how should it be used during resuscitation?

Answer

APGAR evaluates appearance, pulse, grimace, activity, and respiration at one and five minutes after birth. It gives a structured snapshot of condition, but resuscitation decisions should never be delayed to calculate a score; act on heart rate and respiratory effort as they present.

#475 • Special Populations

Why is a small towel or padding placed under an infant's shoulders rather than the head to achieve a neutral airway?

Answer

An infant's head is proportionally larger and heavier relative to body size, which causes passive neck flexion when supine and can obstruct the airway. Padding under the shoulders instead of the head helps achieve a truly neutral position.

#476 • Special Populations

Name three pediatric airway differences from adults that increase the risk of airway complications.

Answer

Children have a proportionally larger tongue relative to the oral cavity, a narrower airway at every level, and a shorter, more flexible trachea, making it easier to over-insert an airway adjunct or kink the airway with improper positioning.

#477 • Special Populations

Why can a normal blood pressure reading in a child be misleading during shock?

Answer

A child's cardiovascular system can maintain a normal blood pressure for a surprisingly long time through tachycardia and peripheral vasoconstriction, so a normal reading does not rule out significant blood loss. Blood pressure drops late and decompensation can then be rapid.

#478 • Special Populations

Which vital signs are more reliable early indicators of pediatric shock than blood pressure?

Answer

Heart rate, capillary refill, skin colour, and mental status are more reliable early indicators of pediatric shock than blood pressure alone.

#479 · Special Populations

What is a febrile seizure and in which age group does it typically occur?

Answer

A febrile seizure occurs in young children, typically between six months and five years of age, during a rapid rise in body temperature. It is usually brief and self-limiting but frightening for caregivers.

#480 · Special Populations

What is croup and how is it generally managed?

Answer

Croup is a viral upper airway infection causing a distinctive barking cough and inspiratory stridor. It is generally managed with cool humidified air and calm positioning, since agitation worsens airway swelling.

#481 • Special Populations

What are the warning signs of a serious upper airway emergency resembling epiglottitis, and what must never be done?

Answer

Warning signs include drooling, a muffled voice, and the tripod position (sitting upright, leaning forward) with rapid deterioration. These children must never be laid flat, have their mouth examined with a tongue depressor, or be separated from a calming caregiver, since agitation can precipitate complete airway obstruction.

#482 • Special Populations

What is bronchiolitis and who does it most commonly affect?

Answer

Bronchiolitis, most often caused by respiratory syncytial virus, affects infants and presents with wheezing, increased work of breathing, and feeding difficulty.

#483 • Special Populations

What signs suggest dehydration in a pediatric patient?

Answer

Signs of pediatric dehydration include dry mucous membranes, a sunken fontanelle in infants, decreased urine output, lethargy, and poor skin turgor.

#484 • Special Populations

What are the three components of the pediatric assessment triangle?

Answer

The pediatric assessment triangle is a rapid, hands-off evaluation built on appearance (tone, interactiveness, consolability, gaze, speech or cry), work of breathing (abnormal sounds, retractions, positioning, flaring), and circulation to skin (pallor, mottling, or cyanosis).

#485 • Special Populations

Why is the pediatric assessment triangle useful before you ever touch the child?

Answer

It lets you form a general impression of how sick a child is within seconds, from across the room, so you can prioritize your next actions, whether an immediate life-saving intervention or a more gradual structured assessment.

#486 • Special Populations

Why is weight-based dosing central to pediatric medication administration?

Answer

A fixed adult dose can be dangerously excessive or ineffective for a child, so medications and fluid resuscitation are dosed by weight. Length-based tools are commonly used in the field to estimate weight when an actual weight is unknown.

#487 • Special Populations

What communication strategies help when assessing a pediatric patient?

Answer

Use simple, honest language, avoid words that can be misinterpreted by a literal-minded child, involve a favourite toy or comfort object when possible, and get down to the child's eye level rather than towering over them.

#488 • Special Populations

How does family-centred care help calm a pediatric patient?

Answer

A calm caregiver is often the paramedic's best tool for keeping a child calm. Whenever safe, allow a parent or guardian to remain close, hold the child, or assist with simple tasks like holding an oxygen mask near the child's face.

#489 • Special Populations

What findings raise concern for non-accidental trauma in a child?

Answer

Warning signs include injuries inconsistent with the stated mechanism, injuries in various stages of healing, patterns suggestive of an object (belt marks, cigarette burns, grip marks), injuries to non-mobile infants, a delay in seeking care, and a history that changes between caregivers or does not match the child's developmental capability.

#490 • Special Populations

What is a paramedic's legal duty in Ontario regarding suspected child abuse or neglect?

Answer

Paramedics have a legal duty to report suspected child abuse or neglect, and this duty exists independently of parental consent or objection. Document objectively what was observed and heard, using direct quotations where appropriate.

#491 • Special Populations

Why might an older adult in shock fail to show the expected tachycardic response?

Answer

Decreased cardiac reserve, stiffened arteries, and blunted baroreceptor response mean older adults may not mount the expected tachycardic response to shock or pain, and beta-blocker medications can mask this further.

#492 • Special Populations

What is a 'silent MI' and why does it matter in geriatric assessment?

Answer

A silent MI is when an older adult experiencing a myocardial infarction presents with vague weakness, confusion, or shortness of breath rather than classic crushing chest pain, so a paramedic must maintain a high index of suspicion even with a mild or nonspecific presentation.

#493 · Special Populations

How might infection present differently in an older adult compared to a younger patient?

Answer

Infection in an older adult may present as new confusion, lethargy, or a fall rather than fever, since the aging immune system mounts a blunted febrile response.

#494 · Special Populations

Why is polypharmacy a concern when assessing an older adult?

Answer

Polypharmacy, the use of multiple medications simultaneously, increases the risk of drug interactions, side effects, and diagnostic confusion, since a medication side effect can mimic an acute illness. A complete medication list, including over-the-counter drugs and supplements, should be obtained.

#495 • Special Populations

Why can pain scores understate the severity of injury in older adults?

Answer

Altered pain perception from neuropathy, cognitive impairment, or chronic pain conditions means pain scores may understate the severity of an underlying injury or illness, so physical findings and history carry extra weight.

#496 • Special Populations

What should a thorough fall assessment in an older adult investigate beyond the injury itself?

Answer

A thorough fall assessment should ask about the events immediately before the fall, since syncope, a cardiac event, medication effects, orthostatic hypotension, or an environmental hazard may all be contributing causes requiring identification and documentation.

#497 • Special Populations

What warning signs during a home visit might suggest elder abuse or neglect?

Answer

Warning signs include unexplained injuries, poor hygiene, an unsafe or unsanitary living environment, medication mismanagement, or a caregiver who is unwilling to leave the patient alone with the paramedic.

#498 • Special Populations

How should a paramedic approach a presented Do Not Resuscitate confirmation form, and what happens in its absence?

Answer

Paramedics must verify and honour a valid Do Not Resuscitate confirmation form or advance directive when one is presented. In its absence, resuscitation is generally initiated.

#499 • Special Populations

How can excess adipose tissue affect airway management in a bariatric patient?

Answer

Excess adipose tissue around the neck, chest, and abdomen can restrict chest wall movement, reduce functional lung capacity, and make bag-valve-mask ventilation and advanced airway insertion more technically difficult.

#500 • Special Populations

Why is safe patient handling planning so important with bariatric patients?

Answer

Standard cots and stretchers have weight limits that can be exceeded, so services carry bariatric-rated equipment. Crews should assess the safe working load of equipment before a lift, request additional resources early, use proper body mechanics and team lifting, and plan egress including doorways, stairwells, and elevator dimensions before the lift begins.